

# Artificial Intelligence in Healthcare Management and Economics

*Evidence, Value, Governance, and Lifecycle Decision-Making*

## Digital Resource Companion

Institutional Records, Workbooks, Worked Examples,  
Reference Guides, and Analytical Resources

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Developed and maintained by the authors

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Printable and editable edition aligned with the four-part, twelve-chapter book.

Worked examples and demonstration datasets are synthetic unless explicitly stated.

# Contents

|                                                                                      |    |
|--------------------------------------------------------------------------------------|----|
| About This Companion.....                                                            | 4  |
| Resource Architecture.....                                                           | 4  |
| Repository Navigation and Current Release.....                                       | 5  |
| Master Decision Lifecycle and Decision Stages.....                                   | 5  |
| Process Maps and Browser-Editable Forms.....                                         | 5  |
| Python Analytics.....                                                                | 5  |
| Consolidated Offline Bundles.....                                                    | 5  |
| Machine-Readable Records and Catalogs.....                                           | 5  |
| How to Use the Companion.....                                                        | 6  |
| Common Institutional Record.....                                                     | 6  |
| Standard Decision Statement.....                                                     | 7  |
| Reader Pathways.....                                                                 | 7  |
| Current Book and Companion Structure.....                                            | 9  |
| File Formats and Repository Structure.....                                           | 9  |
| Cross-Cutting Reference Guides.....                                                  | 10 |
| Version Control, Data Protection, and Citation.....                                  | 11 |
| Status Taxonomy for External and Author-Developed Instruments.....                   | 12 |
| Part I: Managerial and Economic Foundations.....                                     | 12 |
| Chapter 1: AI as a Health System Intervention.....                                   | 12 |
| Chapter 2: Institutional Readiness and Digital Foundations.....                      | 17 |
| Chapter 3: Human–AI Decision Systems and Managerial Action.....                      | 23 |
| Part II: Organizational Performance, Workforce, and Quality.....                     | 28 |
| Chapter 4: Productivity, Work Redesign, and Implementation.....                      | 28 |
| Chapter 5: Quality Management, Accreditation, and Organizational Learning.....       | 33 |
| Part III: Economic Evaluation, Decision Modelling, and Procurement.....              | 38 |
| Chapter 6: Lifecycle Cost, Pricing, and Benefit Realization.....                     | 38 |
| Chapter 7: Economic Evaluation, Affordability, and Health Technology Assessment..... | 43 |
| Chapter 8: Decision Modelling, Simulation, and the Value of Information.....         | 48 |
| Chapter 9: Problem-First Procurement and Vendor Governance.....                      | 64 |
| Part IV: Governance, Evidence, and Lifecycle Decision-Making.....                    | 69 |
| Chapter 10: Governance, Accountability, and Institutional Risk.....                  | 69 |
| Chapter 11: Evidence, Validation, and Real-World Evaluation.....                     | 75 |
| Chapter 12: Decision Architecture and Lifecycle Governance.....                      | 80 |
| Cross-Chapter Resources.....                                                         | 87 |
| Repository Workflow for Institutional Projects.....                                  | 88 |
| Suggested Project Folder.....                                                        | 88 |
| Control Principles.....                                                              | 88 |

|                                       |    |
|---------------------------------------|----|
| Customization and Quality Review..... | 88 |
| Versioning and Citation.....          | 89 |
| Glossary.....                         | 89 |
| Abbreviations and Acronyms.....       | 93 |
| Responsible Use.....                  | 95 |
| References.....                       | 95 |

## About This Companion

The Digital Resource Companion is the applied extension of Artificial Intelligence in Healthcare Management and Economics: Evidence, Value, Governance, and Lifecycle Decision-Making. The book contains the durable scholarly argument. The companion provides the records, workbooks, worked examples, decision pathways, reference guides, and computational materials needed to apply that reasoning to a defined institutional setting.

V1.0 follows the current four-part, twelve-chapter book structure and is synchronized with 29 principal figures, 38 principal tables, 19 core institutional records, 16 worked examples, the common case-analysis method, and 36 Python analytical modules. The detailed chapter resources retain the explanatory content and visual language of the original companion while incorporating the revised decision-state, evidence, economic, procurement, supportability, and lifecycle decision logic.

The companion is intended for institutional analysis, education, research design, documentation, and deliberation. It is not a validated clinical instrument and does not constitute legal, regulatory, procurement, cybersecurity, accreditation, quality-certification, financial, or health technology assessment advice. Local evidence, lawful authority, professional judgement, multidisciplinary review, and accountable approval remain necessary.

## Resource Architecture

| Resource layer                     | Purpose                                                                                                                                               | Best use                                                                                                          |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Worked examples                    | Demonstrate the complete path from problem and comparator through assumptions, analysis, uncertainty, interpretation, and conditional recommendation. | Read before adapting a template or building a local model.                                                        |
| Supplemental resources             | Provide editable specifications, checklists, registers, matrices, protocols, monitoring plans, and decision records.                                  | Use to document the local institutional process and preserve an auditable record.                                 |
| Reference resources                | Clarify terminology, methods, formulas, laws, standards, reporting guidance, appraisal tools, frameworks, and limitations.                            | Use to select the right instrument for the question and to avoid treating status or name recognition as evidence. |
| Core and specialist editable files | Supply DOCX, XLSX, CSV/JSON, Markdown, process-flow, and graphic variants.                                                                            | Select the format that best fits the task while retaining the stable resource identifier and version.             |
| Python analytics                   | Provide scripts, notebooks, structured synthetic data, tests, and outputs for quantitative tasks.                                                     | Use when reproducibility and computational analysis materially improve the decision.                              |

The layers are designed to be used together. A worked example demonstrates the reasoning; an institutional record preserves the local decision; a workbook structures repeated entries and calculations; a reference guide clarifies method and status; and a computational resource supports reproducible analysis. No single file replaces the living decision record.

## Repository Navigation and Current Release

The public release is presented through four complementary routes. The GitHub Pages Web Companion is the principal browser-based entry point. The Reader and Practitioner Edition is the recommended downloadable package for most users. The Core Tools Pack provides the consolidated institutional forms and workbook, completion guidance, licence terms, and a manifest in a lightweight collection. The Canonical Source contains the complete authoritative repository used to maintain, reproduce, and publish the companion.

### Master Decision Lifecycle and Decision Stages

A ten-stage Master Institutional Decision Lifecycle links problem definition, readiness, evidence, economics, modelling, governance, procurement, implementation, continuing value, and renewal or retirement. Readers can browse by chapter when they already know the subject area or by decision stage when a project spans several chapters. Each stage identifies recommended preceding records, primary and supporting resources, a typical output, and a decision gate. The sequence is iterative rather than linear: validation findings, procurement constraints, incidents, material changes, or new alternatives may require a return to an earlier stage.

### Process Maps and Browser-Editable Forms

The GitHub Pages layer presents the master lifecycle, twelve-chapter process maps, resource-level process maps, browser-readable versions of the 19 core institutional records and seven cross-cutting reference guides, a searchable catalogue, sixteen worked examples, and selected interactive demonstrations. Browser forms are drafting and learning aids. Their contents should be transferred to governed institutional records before consequential use.

### Python Analytics

The Python Analytics collection contains 36 chapter-aligned modules. The modules address data quality, validation, calibration, decision utility, lifecycle cost, financial recognition, cost-effectiveness, budget impact, sensitivity analysis, simulation, causal evaluation, monitoring, decision provenance, lifecycle decision-state and trigger review. Method choice should follow the decision problem, and consequential outputs require independent checking, versioned inputs, preserved code and environments, and a static decision record.

### Consolidated Offline Bundles

The consolidated Core Resource Forms and Core Resource Workbooks support committees, workshops, and teams working offline. The standalone Core Tools Pack combines the consolidated forms, consolidated workbook, completion guidance, licence terms, limited-use notice, version information, manifest, and checksums. These are convenience distributions: the chapter and resource folders in the Canonical Source remain the governing source for resource-specific content.

### Machine-Readable Records and Catalogs

Core resources include blank and illustrative structured records, while synchronized catalogues describe resources, figures, tables, worked cases, Python modules, and relationships. Machine-readable files support controlled interchange, validation, and future software integration, but they do not replace human-readable evidence, multidisciplinary review, accountable authority, or institutional governance.

## How to Use the Companion

1. Begin with the institutional decision. Define the population, setting, accountable decision-maker, intended and excluded uses, comparator, time horizon, and consequence of error before selecting a file.
2. Read the aligned book chapter. The book remains the controlling source for conceptual claims, chapter structure, and the interpretation of the author-developed framework package.
3. Review the worked example. Identify the pathway from problem and comparator to evidence, cost, outcome, uncertainty, safeguards, and recommendation.
4. Select the smallest sufficient resource set. Avoid completing every form merely because it exists. Choose the records needed for the consequence, uncertainty, scale, and institutional dependence of the decision.
5. Create a governed local copy. Preserve the original resource identifier and release version, record the local owner, and work in an approved institutional location rather than the public repository.
6. Replace synthetic assumptions. Use current, authorized, versioned evidence. Mark unknowns explicitly and do not represent missing evidence as a neutral or average value.
7. Complete multidisciplinary review. Clinical, operational, patient, technical, economic, quality, privacy, cybersecurity, procurement, legal, and governance input should be proportionate to the decision.
8. Separate observations, modelled estimates, value judgements, and contractual claims. This distinction improves traceability and prevents unsupported certainty.
9. End with an accountable decision. Record the selected and rejected options, conditions, indicators, thresholds, owners, review date, trigger events, and exit route.
10. Maintain the record after deployment. Update the dossier after incidents, material changes, new evidence, price changes, altered workflows, renewal, restriction, replacement, or retirement.

## Common Institutional Record

The following fields apply across the chapter resources and are presented once here to avoid repeating the same generic template in every chapter.

| Field                             | Minimum entry                                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Decision and authority            | Decision to be made; accountable sponsor; decision owner; approval, escalation, and stopping authority.                          |
| Population and setting            | Included and excluded populations, sites, services, users, and time period.                                                      |
| Problem, baseline, and comparator | Current pathway, burden, variation, unmet need, and all realistic AI and non-AI alternatives.                                    |
| Intervention boundary             | Data, model or method, interface, people, workflow, capacity, implementation, monitoring, vendor support, and exit.              |
| Evidence and version              | Sources, dates, exact system and software version, applicability, confidence, gaps, assumptions, and evidence cut-off.           |
| Costs and consequences            | Perspective, budget holder, resource use, outcomes, workload, distribution, opportunity cost, and residual value where relevant. |

| Field                  | Minimum entry                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Mandatory safeguards   | Patient safety, lawful data use, privacy, cybersecurity, equity, accessibility, meaningful oversight, continuity, and contestability. |
| Decision conditions    | Proceed, redesign, restrict, pilot, defer, replace, or retire criteria; milestones; contractual duties; and review date.              |
| Monitoring and closure | Indicators, thresholds, triggers, responsible owners, material-change route, renewal decision, transition, data closure, and lessons. |

## Standard Decision Statement

For the specified decision problem and comparator set, the institution should [proceed / proceed with conditions / redesign / pilot / restrict / defer / replace / retire]. The decision applies to [population, setting, system and version, and evidence period], is owned by [accountable authority], and will be reviewed on [date] or earlier if [trigger events] occur. The evidence, assumptions, unresolved uncertainty, mandatory safeguards, monitoring thresholds, material-change route, and exit arrangements are recorded in the accompanying dossier.

## Reader Pathways

The reader-facing navigation begins with five goals: understand the approach, make an institutional decision, use a template or workbook, run an analysis, and learn through a worked case.

Resources are classified as introductory, applied, advanced, or technical. The Web Companion emphasizes core records, worked cases, the common case-analysis method, and reference guides; machine-readable files and analytical modules remain available through the advanced view.

The Reader and Practitioner Edition provides the guided entry files, pathways, Core Resource Forms and Workbooks, sixteen worked examples, reference guides, analytical guidance and browser demonstrations, current catalogues, the printable companion, and an offline Web Companion without exposing the complete development source tree. The GitHub repository remains the Canonical Source for maintenance, reproducibility, integration, publication, and future releases.

| Reader group                                                                         | Suggested route                                                              | Principal use                                                                                                                                                  |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hospital and health system boards, executives, and senior managers                   | Begin with Chapters 1, 6, 9, 10, and 12; use Chapters 2, 3, 4, 5 as relevant | Strategic investment, opportunity cost, capacity and benefit realization, quality and governance oversight, procurement, renewal, replacement, and retirement. |
| Clinical, nursing, allied health, quality, patient safety, and accreditation leaders | Chapters 3, 4, 5, 10, 11, 12                                                 | Human oversight, workflow fit, safety, quality management, accreditation evidence, workforce capability, corrective action, and organizational learning.       |

| Reader group                                                                                               | Suggested route                                                         | Principal use                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Health economists, HTA specialists, outcomes researchers, finance teams, payers, and commissioners         | Chapters 1, 6, 7, 8, 9, followed by Chapters 11, 12                     | Lifecycle costing, pricing, financial recognition, cost-effectiveness, budget impact, affordability, value of information, distribution, uncertainty, and lifecycle reassessment.       |
| Digital health, health informatics, data science, and implementation teams                                 | Chapters 2, 3, 4, 8, 9, 10, 11, 12                                      | Readiness, interoperability, data provenance, local validation, workflow integration, configuration control, monitoring, material change, and reproducibility.                          |
| Ethics, legal, privacy, cybersecurity, data governance, patient safety, and institutional assurance bodies | Chapters 2, 3, 9, 10, 11, 12                                            | Rights, lawful use, accountability, contestability, privacy, cybersecurity, equity, continuity, trust, incident response, and stopping authority.                                       |
| Procurement, supply chain, biomedical engineering, and asset management                                    | Chapters 6, 9, and 12                                                   | Pricing, disclosure, contracting, supportability, infrastructure dependencies, maintenance, replacement, residual value, renewal, transition, and exit.                                 |
| Policymakers, regulators, public health authorities, accreditation organizations, and standards bodies     | Chapters 1, 2, 5, 7, and 9–12                                           | Readiness, public value, quality and accreditation, HTA, assurance context, evidence requirements, responsible scaling, and lifecycle oversight.                                        |
| Patient and public representatives                                                                         | Chapters 1, 3, 7, 10, and 12                                            | Participation, transparency, meaningful oversight, effective access, equity, contestability, benefits and burdens, and public value.                                                    |
| Researchers, educators, students, and practitioners                                                        | Chapters 1, 7, 8, and 10–12, with the chapters relevant to the question | Research design, appraisal, causal evaluation, economic and decision modelling, governance analysis, reproducibility, teaching, and applied use.                                        |
| Technology developers and suppliers                                                                        | Chapters 1, 2, 3, 4, 5, 6, 9, 10, 11, 12                                | Institutional expectations for problem definition, evidence, interoperability, integration, oversight, quality, pricing, procurement, continuing disclosure, material change, and exit. |

## Current Book and Companion Structure

| Part     | Title                                                    | Aligned chapters                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part I   | Managerial and Economic Foundations                      | Chapter 1: AI as a Health System Intervention; Chapter 2: Institutional Readiness and Digital Foundations; Chapter 3: Human–AI Decision Systems and Managerial Action                                                                                                             |
| Part II  | Organizational Performance, Workforce, and Quality       | Chapter 4: Productivity, Work Redesign, and Implementation; Chapter 5: Quality Management, Accreditation, and Organizational Learning                                                                                                                                             |
| Part III | Economic Evaluation, Decision Modelling, and Procurement | Chapter 6: Lifecycle Cost, Pricing, and Benefit Realization; Chapter 7: Economic Evaluation, Affordability, and Health Technology Assessment; Chapter 8: Decision Modelling, Simulation, and the Value of Information; Chapter 9: Problem-First Procurement and Vendor Governance |
| Part IV  | Governance, Evidence, and Lifecycle Decision-Making      | Chapter 10: Governance, Accountability, and Institutional Risk; Chapter 11: Evidence, Validation, and Real-World Evaluation; Chapter 12: Decision Architecture and Lifecycle Governance                                                                                           |

## File Formats and Repository Structure

| Format | Typical purpose                                                                                                  | Selection guidance                                                                              |
|--------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| DOCX   | Narrative records, committee papers, protocols, assurance notes, local guidance, and editable publication files. | Use when explanation, review, sign-off, and versioned prose are central.                        |
| XLSX   | Registers, costing, repeated records, scenarios, comparisons, dashboards, and selected worked examples.          | Use when rows are updated over time, formulas are required, or structured comparison is needed. |
| PDF    | Stable readable and printable reference copies.                                                                  | Use for distribution and review; retain the corresponding editable source for local adaptation. |

| Format                         | Typical purpose                                                                                     | Selection guidance                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| CSV / JSON                     | Structured exchange, audit trails, schemas, synthetic data, catalogues, and reproducible workflows. | Use for controlled interchange and machine-readable records; keep a human-readable decision record alongside them. |
| Markdown / Mermaid / SVG / PNG | Repository documentation, process descriptions, editable diagrams, and publication-ready visuals.   | Use for transparent version-controlled guidance and reproducible process maps.                                     |

A typical repository path is:

digital-resource-companion/

├─ README.md, GETTING\_STARTED.md, BOOK\_STRUCTURE.md, and RESOURCE\_INDEX.md

├─ chapters/ Core records, guides, analytical modules, and chapter process maps

├─ worked\_examples/ Canonical worked cases and the common case-analysis method

├─ process/ Master lifecycle, decision stages, and dependency maps

├─ bundles/ Consolidated forms and workbooks

├─ publications/ Definitive printable companion

├─ catalogs/ Resource, case, analytical, relationship, and figure catalogs

└─ docs/ GitHub Pages website

The stable resource identifier, not the file path alone, should be recorded in institutional work. Folders and filenames may change between releases; the resource identifier, version, title, and release date provide the traceable citation.

## Cross-Cutting Reference Guides

The cross-cutting guides address questions that recur across chapters and are especially likely to change as law, standards, reporting guidance, software practice, and institutional policy evolve.

| ID        | Guide                                        | Purpose                                                                                                                                                                                                    | Primary alignment                    |
|-----------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| AIHE-RG01 | Framework Status and Selection Guide         | Distinguish legal instruments, standards, guidance, reporting guidelines, appraisal tools, implementation frameworks, and author-developed decision aids; select them according to function and authority. | Chapter 12; aligned 2, 4, 5, 7, 9-12 |
| AIHE-RG02 | Reporting and Appraisal Tool Selection Guide | Match evidence type and study stage to current reporting and appraisal instruments without treating checklist completion as proof of validity, effectiveness, or regulatory conformity.                    | Chapter 11; aligned 7, 8, 11         |

| ID        | Guide                                                  | Purpose                                                                                                                                                                           | Primary alignment                         |
|-----------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| AIHE-RG03 | Implementation Framework Selection Guide               | Select CFIR, NASSS-CAT, RE-AIM/PRISM, TAM/UTAUT, and mixed-method approaches according to the implementation question rather than name recognition.                               | Chapter 4; aligned 4, 11, 12              |
| AIHE-RG04 | Regulatory, Standards, and Policy Update Register      | Maintain date-stamped records for European, Romanian, and global legal, policy, standards, reimbursement, and guidance sources whose scope or implementation status may change.   | Chapter 10; aligned 2, 5, 7, 9, 10, 12-15 |
| AIHE-RG05 | Scenario Planning and Real-Options Worksheet           | Examine uncertainty through alternative futures, thresholds, staged commitments, learning options, contract flexibility, and explicit exit criteria.                              | Chapter 8; aligned 1, 6-9, 12, 15         |
| AIHE-RG06 | Diagnostic and Predictive Evidence Appraisal Checklist | Appraise reference standards, index tests, data partitions, calibration, diagnostic accuracy, bias, subgroup performance, clinical utility, and downstream resource consequences. | Chapter 11; aligned 3, 7, 8, 11           |
| AIHE-RG07 | Common AI Adoption Failures and Corrective Actions     | Identify recurring technology-led, economic, workflow, evidence, governance, procurement, and lifecycle failures and record preventive controls.                                  | Chapter 1; aligned 1-12                   |

A guide organizes selection and documentation; it does not certify compliance, methodological quality, safety, effectiveness, cost-effectiveness, or readiness.

## Version Control, Data Protection, and Citation

- Download or clone a versioned release and retain the unmodified source copy.
- Record the resource identifier, title, version, release date, date accessed, and local modification date.
- Maintain a change log identifying the field or formula changed, the reason, the owner, the evidence source, and whether reapproval is required.

- Record the exact model, software, data, interface, prompt, threshold, and workflow version when they affect the decision.
- Do not enter identifiable patient information, credentials, private keys, confidential contracts, or restricted institutional data into public or uncontrolled copies.
- Store completed resources according to institutional retention, access, audit, and incident-response requirements.
- Cite both the book and the specific companion release used. A local adaptation should be described as an adaptation rather than represented as the unchanged published resource.
- GitHub supports the active working repository; Zenodo or another archival repository should hold citable, immutable releases. Use the DOI and repository URL shown on the release page.

## Status Taxonomy for External and Author-Developed Instruments

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- Law or regulation: establishes binding obligations within its jurisdiction and scope.
- Standard or management system: structures repeatable practices or requirements but does not replace legal interpretation or local evidence.
- Reporting guideline: improves transparency of a report; completion does not prove methodological validity or effectiveness.
- Appraisal tool: supports critical review of bias and applicability; it does not make the institutional decision.
- Implementation framework or theory: organizes determinants, processes, outcomes, context, or acceptance questions.
- Data or interoperability specification: supports exchange or analysis but does not establish semantic consistency, data fitness, or readiness by itself.
- Author-developed framework or process: a proposed decision aid requiring explicit status, versioning, limitations, and independent evaluation before standardized use.

Select each instrument for a defined question, document why it was chosen, and verify its current edition and jurisdictional status before use.

## Part I: Managerial and Economic Foundations

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This Part establishes the managerial and economic problem, the institutional foundations required for action, and the design of human–AI decision support.

### Chapter 1: AI as a Health System Intervention

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#### Chapter Orientation

Chapter 1 establishes the companion’s governing unit of analysis: the complete health-system intervention rather than the model alone. The resources help a team define the service problem, compare AI and non-AI alternatives, identify the causal path from information to action and value, and preserve the ability to change or stop the decision.

Use this suite at project intake, portfolio prioritization, and before market engagement. It is especially useful when a proposal begins with a product claim rather than a documented problem, or when different departments are using inconsistent assumptions about scope, comparator, outcomes, ownership, and opportunity cost.

## Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                          | How to use it                                                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Emergency-Triage Intervention Portfolio                                                   | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Intervention Specification; Comparator and Opportunity-Cost Record; Decision Conditions and Review Record | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | Intervention bundle, Comparator, Public value                                                             | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |

## Worked Example: Emergency-Triage Intervention Portfolio

### *Purpose and Status*

A regional public hospital is considering an AI-supported emergency-triage service after repeated crowding, prolonged time to first clinical assessment, and concern about delayed recognition of deteriorating patients. The proposal must be assessed as a complete intervention rather than as a prediction model.

### *Decision Problem and Comparators*

Decision problem. Should the hospital pilot an AI-supported triage pathway, strengthen the current pathway through staffing and protocol redesign, or adopt a lower-complexity rules-based alternative?

Table: Comparators for the Chapter 1 worked example

| Option | Comparator or strategy                              |
|--------|-----------------------------------------------------|
| 1      | Current triage with existing staffing               |
| 2      | Additional triage staffing during peak periods      |
| 3      | Registration and streaming redesign                 |
| 4      | A conventional rules-based early-warning score      |
| 5      | AI-supported prioritization with human confirmation |

### *Synthetic Parameters and Evidence Sources*

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 1 worked example

| Parameter                                      | Base value      | Source or interpretation                       |
|------------------------------------------------|-----------------|------------------------------------------------|
| Annual eligible attendances                    | 72,000          | Local activity report                          |
| Baseline median time to first assessment       | 84 minutes      | Emergency-department dashboard                 |
| High-risk event prevalence                     | 4.5%            | Retrospective cohort                           |
| AI sensitivity at proposed threshold           | 0.88            | External evidence; local verification required |
| AI specificity at proposed threshold           | 0.72            | External evidence; local verification required |
| Incremental annual programme cost              | EUR 310,000     | Local lifecycle-cost estimate                  |
| Available additional rapid-assessment capacity | 18 patients/day | Workforce and space assessment                 |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                                                          | Decision record                               |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Map the present pathway and identify the decision window in which prioritization could still change care.                                              | Evidence and responsible owner to be recorded |
| 2    | Compare all realistic alternatives before assigning value to the AI option.                                                                            | Evidence and responsible owner to be recorded |
| 3    | Represent the intervention bundle: data availability, interface, triage-nurse review, escalation, rapid-assessment capacity, monitoring, and fallback. | Evidence and responsible owner to be recorded |
| 4    | Estimate false-positive workload and false-negative consequences at the proposed threshold.                                                            | Evidence and responsible owner to be recorded |
| 5    | Test whether downstream capacity can absorb the additional patients identified as high risk.                                                           | Evidence and responsible owner to be recorded |
| 6    | Define a reversible pilot with clinical, operational, economic, equity, and trust indicators.                                                          | Evidence and responsible owner to be recorded |

## Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                                     | Result                                   | Interpretation                                                                                               |
|--------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Expected high-risk alerts per day          | Approximately 61                         | $72,000/365 \times [0.045 \times 0.88 + 0.955 \times (1 - 0.72)]$ ; includes true and false positive alerts. |
| Additional patients requiring rapid review | Approximately 35/day                     | Illustrative incremental-workload assumption; exceeds the stated capacity of 18 additional patients/day.     |
| Primary value mechanism                    | Earlier review of high-risk patients     | Dependent on response capacity                                                                               |
| Main economic risk                         | Unusable alerts and displaced staff time | May increase rather than reduce delay                                                                        |
| Preferred initial decision                 | Redesign and limited pilot               | Not unrestricted deployment                                                                                  |

## Interpretation and Recommendation

Proceed only with a redesigned, time-limited pilot that adds response capacity and compares the AI pathway with a staffing-and-streaming alternative. The model should remain advisory, with triage-nurse confirmation and a documented fallback.

- complete local calibration and silent-mode evaluation.
- demonstrate that rapid-assessment capacity can absorb expected alert volume.
- define subgroup and balancing measures, including delay among patients deprioritised by the system.
- cap total pilot expenditure and preserve the ability to stop without service disruption.
- review at 8 and 16 weeks using predefined proceed, redesign, restrict, or stop criteria.

## Supplemental Resources

These supplemental resources translate the Chapter 1 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 1 supplemental-resource set

| Resource                               | Purpose and minimum content                                                                              | Expected output                                       |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Intervention Specification             | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Comparator and Opportunity-Cost Record | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

| Resource                              | Purpose and minimum content                                                                              | Expected output                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Decision Conditions and Review Record | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Intervention Specification*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Comparator and Opportunity-Cost Record*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Map the present pathway and identify the decision window in which prioritization could still change care. - Compare all realistic alternatives before assigning value to the AI option. - Represent the intervention bundle: data availability, interface, triage-nurse review, escalation, rapid-assessment capacity, monitoring, and fallback. - Estimate false-positive workload and false-negative consequences at the proposed threshold. - Test whether downstream capacity can absorb the additional patients identified as high risk. - Define a reversible pilot with clinical, operational, economic, equity, and trust indicators.

### *Decision Conditions and Review Record*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### *Key Terms*

Table: Key terms for Chapter 1

| Term                | Working definition and decision relevance                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Intervention bundle | Model, data, interface, people, workflow, capacity, governance, monitoring, and exit.               |
| Comparator          | The best realistic alternative use of the same resources.                                           |
| Public value        | Clinical, economic, organizational, equity, legitimacy, and stewardship effects considered jointly. |

### *Status and Appropriate Use of Selected Instruments*

Table: Selected instruments relevant to Chapter 1

| Instrument                 | Status or type     | Appropriate use and limitation                              |
|----------------------------|--------------------|-------------------------------------------------------------|
| Health-system intervention | Analytical framing | Defines the complete unit of analysis.                      |
| Value chain                | Causal map         | Shows where capability may be converted into or lose value. |

| Instrument          | Status or type | Appropriate use and limitation                        |
|---------------------|----------------|-------------------------------------------------------|
| Economic evaluation | Method family  | Compares costs and outcomes relative to a comparator. |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                      | Purpose                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-A01 — Decision and Use-Case Specification<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                              | Define the service problem, full intervention, realistic comparators, outcomes, value mechanism, decision rule, and review cycle.                |
| AIHE-A02 — Stakeholders, Decision Rights, and Participation<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                 | Identify affected parties, decision rights, participation duties, evidence contributions, and escalation authority.                              |
| AIHE-CASE00 — Common Case-Analysis Method<br>Case-analysis method; md; docx; xlsx; csv; json                                  | Provide a consistent problem-first, evidence, economic, governance, and lifecycle method for analysing worked and local cases.                   |
| AIHE-CASE01 — Emergency-Department Triage Portfolio Decision<br>Worked case; md; csv; json; xlsx                              | Which intervention should the hospital test first when AI, staffing, and process redesign compete for the same limited resources?                |
| AIHE-RG07 — Common AI Adoption Failures and Corrective Actions<br>Reference guide; docx; xlsx; csv; json; md                  | Identify recurring technology-led, economic, workflow, evidence, governance, procurement, and lifecycle failures and record preventive controls. |
| AIHE-EX01 — Project Intake and Stakeholder Register<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Capture the initiating problem, project scope, accountable owners, affected groups, and decision rights.                                         |
| AIHE-W — AI Use-Case Canvas<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                         | Define the problem, population, intended users, intervention bundle, comparator, outcomes, and initial risks.                                    |
| AIHE-X — Decision Problem Template<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                  | Specify population, intervention, comparator, outcomes, perspective, setting, time horizon, and decision rule.                                   |

## Chapter 2: Institutional Readiness and Digital Foundations

### Chapter Orientation

Chapter 2 addresses the foundations that determine whether an AI-enabled service can operate reliably: data quality, interoperability, infrastructure, resilience, stewardship, and institutional capability. The resources distinguish general digital maturity from readiness for a specific use case and make shared foundation investments visible as portfolio assets.

Use this suite before local validation or procurement when source data, identity matching, terminology, interfaces, provenance, latency, security, or workflow ownership may constrain the proposed intervention. The output should

show what is ready, what requires remediation, what can be tested safely, and which investments can be reused across future services.

### Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                                 | How to use it                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Readmission Analytics Readiness and Data-Foundation Investment                                   | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Use-Case Readiness Profile; Data and Interoperability Evidence Record; Foundation Investment and Portfolio Reuse | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | FHIR, OMOP, Digital maturity                                                                                     | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |
| Repository coverage    | 2 core, 3 python analytics, 5 supplemental specialist tool, 1 worked case                                        | Use the catalog map below to locate editable, process-flow, data, and computational variants.                  |

### Worked Example: Readmission Analytics Readiness and Data-Foundation Investment

#### *Purpose and Status*

A multi-site hospital group proposes a readmission-risk service, but discharge data, medication reconciliation, laboratory results, and prior admissions are distributed across several systems. The decision is whether to procure a model immediately or first strengthen reusable digital foundations.

#### *Decision Problem and Comparators*

Decision problem. Which sequence of investment offers the greatest institutional value: immediate procurement, a foundation-first programme, or a limited proof of concept restricted to one service?

Table: Comparators for the Chapter 2 worked example

| Option | Comparator or strategy                                                                       |
|--------|----------------------------------------------------------------------------------------------|
| 1      | Immediate vendor procurement                                                                 |
| 2      | Foundation-first investment in identity, terminology, interfaces, provenance, and monitoring |
| 3      | Single-site proof of concept with manual data preparation                                    |
| 4      | Conventional discharge-risk checklist and additional care-coordination capacity              |

#### *Synthetic Parameters and Evidence Sources*

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 2 worked example

| Parameter                                             | Base value  | Source or interpretation   |
|-------------------------------------------------------|-------------|----------------------------|
| Hospitals in scope                                    | 4           | Portfolio plan             |
| Interfaces requiring remediation                      | 11          | Architecture inventory     |
| Records with reliable medication reconciliation       | 68%         | Data-quality profile       |
| Cross-site patient-identity match rate                | 96.1%       | Master-patient-index audit |
| Estimated foundation investment                       | EUR 740,000 | Programme estimate         |
| Use cases expected to reuse the foundations           | 7           | Digital portfolio          |
| Estimated procurement delay from foundation programme | 9 months    | Implementation plan        |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                              | Decision record                               |
|------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Specify the data, timing, and workflow dependencies of the intended use.                                   | Evidence and responsible owner to be recorded |
| 2    | Assess readiness by use case rather than by one maturity score.                                            | Evidence and responsible owner to be recorded |
| 3    | Identify which weaknesses affect several proposed services and estimate their portfolio-level reuse value. | Evidence and responsible owner to be recorded |
| 4    | Compare immediate procurement with a foundation-first sequence and a restricted proof of concept.          | Evidence and responsible owner to be recorded |
| 5    | Define acceptance criteria for identity, semantics, provenance, latency, resilience, and governance.       | Evidence and responsible owner to be recorded |
| 6    | Record which standards are implementation tools and which local mappings and controls remain necessary.    | Evidence and responsible owner to be recorded |

## Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                            | Result                                            | Interpretation                                        |
|-----------------------------------|---------------------------------------------------|-------------------------------------------------------|
| Immediate procurement feasibility | Low                                               | Critical data and identity dependencies unresolved    |
| Foundation reuse potential        | High                                              | Seven planned services share the same capabilities    |
| Short-term delay cost             | Moderate                                          | Potential benefit deferred for nine months            |
| Risk of isolated procurement      | High                                              | Repeated mapping, weak provenance, and technical debt |
| Preferred decision                | Foundation-first with restricted proof of concept | Allows learning without operational dependence        |

## Interpretation and Recommendation

Fund the shared digital-foundation programme and permit a restricted, non-clinically dependent proof of concept in one service. Procurement for routine use should wait until predefined data, interoperability, resilience, and governance thresholds are met.

- identity and duplicate-record controls meet the agreed threshold.
- medication and discharge data have documented provenance and completeness.
- interfaces meet decision-latency and recovery requirements.
- local FHIR profiles or equivalent exchange specifications and analytical transformations are version controlled.
- the investment case identifies portfolio reuse and avoids assigning all shared cost to one application.

## Supplemental Resources

These supplemental resources translate the Chapter 2 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 2 supplemental-resource set

| Resource                                  | Purpose and minimum content                                                                              | Expected output                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Use-Case Readiness Profile                | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Data and Interoperability Evidence Record | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

| Resource                                  | Purpose and minimum content                                                                              | Expected output                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Foundation Investment and Portfolio Reuse | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Use-Case Readiness Profile*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Data and Interoperability Evidence Record*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Specify the data, timing, and workflow dependencies of the intended use. - Assess readiness by use case rather than by one maturity score. - Identify which weaknesses affect several proposed services and estimate their portfolio-level reuse value. - Compare immediate procurement with a foundation-first sequence and a restricted proof of concept. - Define acceptance criteria for identity, semantics, provenance, latency, resilience, and governance. - Record which standards are implementation tools and which local mappings and controls remain necessary.

### *Foundation Investment and Portfolio Reuse*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## **Reference Resources**

### *Key Terms*

Table: Key terms for Chapter 2

| Term             | Working definition and decision relevance                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| FHIR             | A standard for structured exchange of health information; it does not remove local profiling, mapping, or governance. |
| OMOP             | A common data model for observational analysis; it is complementary to exchange standards.                            |
| Digital maturity | A profile of capabilities tied to a use case, not a certificate of universal readiness.                               |

### *Status and Appropriate Use of Selected Instruments*

Table: Selected instruments relevant to Chapter 2

| Instrument            | Status or type  | Appropriate use and limitation                                              |
|-----------------------|-----------------|-----------------------------------------------------------------------------|
| HIMSS maturity models | Maturity models | Road maps for capability development; not proof of readiness for every use. |

| Instrument | Status or type                                   | Appropriate use and limitation                                                       |
|------------|--------------------------------------------------|--------------------------------------------------------------------------------------|
| FHIR       | Interoperability standard                        | Supports exchange; requires local profiles and conformance.                          |
| OMOP CDM   | Analytical data model                            | Supports observational analysis; does not replace source-system governance.          |
| EHDS       | Binding EU regulation with phased implementation | Shapes primary and secondary use, interoperability, access, and governance in scope. |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                        | Purpose                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-A03 — Institutional Readiness Assessment Core; docx; xlsx; csv; json; md; mmd; svg; png                                    | Assess strategic, data, infrastructure, workflow, workforce, quality, economic, governance, procurement, and continuing support readiness.                  |
| AIHE-A04 — Data Source, Provenance, and Fitness Record Core; docx; xlsx; csv; json; md; mmd; svg; png                           | Document source and derived data assets, authority, definitions, transformations, quality, lineage, monitoring, and remediation.                            |
| AIHE-CASE02 — Readmission-Model Readiness and Data-Fitness Assessment Worked case; md; csv; json; xlsx                          | Is the organization ready to validate and pilot a readmission-risk model, and which capability gaps must be remediated first?                               |
| AIHE-A — AI Readiness Assessment for Health Systems Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png      | Assess strategic, data, infrastructure, workforce, governance, economic, ethical, security, and monitoring readiness at organizational and use-case levels. |
| AIHE-EX03 — Data Source Map and Access Record Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png            | Document data sources, access authority, interoperability, quality, provenance, and reuse constraints.                                                      |
| AIHE-L — Health-System AI Readiness Maturity Assessment Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png  | Rate health-system maturity by domain and identify the weakest critical capability and prerequisites before pilot or scale.                                 |
| AIHE-U — SHARE-AI Secondary-Use Governance Worksheet Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png     | Authorize secondary data use through purpose, harmonization, access, risk, public value, and audit controls.                                                |
| AIHE-Y — Data Quality and Fitness-for-Purpose Assessment Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Assess completeness, plausibility, consistency, timeliness, representativeness, lineage, and actionability.                                                 |

| Resource                                                                                                                  | Purpose                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-PY01 — Data quality, fitness-for-purpose, and mapping audit<br>Python analytics; py; ipynb; xlsx; csv; json; png; md | Profiles missingness, duplicates, implausible values, subgroup coverage, and terminology-mapping completeness before AI validation or economic analysis. |
| AIHE-PY03 — Missing-data sensitivity analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                    | Shows how complete-case analysis, simple imputation, and best/worst-case assumptions can change estimated outcomes and costs.                            |
| AIHE-PY33 — Patient or service segmentation using clustering<br>Python analytics; py; ipynb; xlsx; csv; json; png; md     | Uses standardized features and k-means clustering to identify groups with different utilization, risk, cost, or workflow patterns.                       |

## Chapter 3: Human-AI Decision Systems and Managerial Action

### Chapter Orientation

Chapter 3 focuses on the conversion of predictions, recommendations, alerts, and dashboard signals into accountable action. The resources specify decision windows, thresholds, users, human oversight, escalation, uncertainty, action latency, and the balancing measures required to detect overload or displaced burden.

Use this suite when designing clinical decision support, operational dashboards, command centres, or partially automated workflows. It helps prevent technically accurate information from being mistaken for decision value when users lack time, authority, context, or downstream capacity to act.

### Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                                      | How to use it                                                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Hospital Command Dashboard and Patient Flow                                                           | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Human-AI Decision-Support Specification; Dashboard Signal-to-Action Record; Delegation, Oversight, and Action Latency | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | Decision support, Delegation, Action latency                                                                          | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |
| Repository coverage    | 1 core, 4 python analytics, 1 supplemental specialist tool, 2 worked case                                             | Use the catalog map below to locate editable, process-flow, data, and computational variants.                  |

## Worked Example: Hospital Command Dashboard and Patient Flow

### Purpose and Status

A tertiary hospital plans a command dashboard combining admissions, discharges, bed status, staffing, diagnostics, transport, and forecast demand. Leaders expect faster decisions, but current indicators have inconsistent definitions and no explicit action owners.

### Decision Problem and Comparators

Decision problem. Should the hospital implement an AI-enhanced command dashboard, redesign the existing operational review process, or combine both through a staged signal-to-action intervention?

Table: Comparators for the Chapter 3 worked example

| Option | Comparator or strategy                                                      |
|--------|-----------------------------------------------------------------------------|
| 1      | Current morning operational meeting and static reports                      |
| 2      | Standardized dashboard without predictive functions                         |
| 3      | AI-enhanced dashboard with demand forecasts and anomaly detection           |
| 4      | Targeted process redesign for discharge, imaging, and transport bottlenecks |

### Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 3 worked example

| Parameter                               | Base value  | Source or interpretation |
|-----------------------------------------|-------------|--------------------------|
| Average daily admissions                | 312         | Operations data          |
| Mean bed occupancy                      | 94%         | Capacity dashboard       |
| Median data latency                     | 95 minutes  | Interface audit          |
| Indicators without named action owner   | 42%         | Dashboard inventory      |
| Forecast mean absolute percentage error | 13%         | Retrospective validation |
| Estimated annual dashboard cost         | EUR 265,000 | Lifecycle-cost estimate  |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                                | Decision record                               |
|------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Define the operational decisions, decision windows, and accountable owners before selecting indicators.                      | Evidence and responsible owner to be recorded |
| 2    | Separate observed values, model estimates, forecasts, and recommendations.                                                   | Evidence and responsible owner to be recorded |
| 3    | Create an indicator dictionary with numerator, denominator, exclusions, latency, and limitations.                            | Evidence and responsible owner to be recorded |
| 4    | Map each signal to a feasible action and at least one balancing measure.                                                     | Evidence and responsible owner to be recorded |
| 5    | Test forecast value against a standardized non-predictive dashboard and targeted process redesign.                           | Evidence and responsible owner to be recorded |
| 6    | Measure action latency, downstream congestion, workload, and the consequences of recommendations that cannot be implemented. | Evidence and responsible owner to be recorded |

### *Illustrative Outputs*

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                              | Result                                          | Interpretation                                 |
|-------------------------------------|-------------------------------------------------|------------------------------------------------|
| Principal failure in current design | No action ownership                             | Information does not reliably change decisions |
| Forecast technical performance      | Potentially useful                              | Requires local operational validation          |
| Main value constraint               | High occupancy and limited downstream capacity  | Forecasting cannot create beds or staff        |
| Preferred intervention              | Signal-to-action redesign plus staged dashboard | Dashboard alone is insufficient                |
| Primary monitoring outcome          | Decision and action latency                     | Views and logins are secondary                 |

### *Interpretation and Recommendation*

Implement a standardized operational dashboard first, with explicit action ownership and decision thresholds. Add forecasting functions only after the data-latency and response-pathway requirements have been met and tested in simulation and a controlled pilot.

- all priority indicators have definitions, owners, decision windows, and balancing measures.
- data latency remains within the operational window.
- forecast uncertainty is displayed and observed values are distinguishable from predictions.
- the pilot measures action and outcome changes rather than views or user satisfaction alone.
- a governance process reviews gaming, surveillance concerns, and unequal workload effects.

## Supplemental Resources

These supplemental resources translate the Chapter 3 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 3 supplemental-resource set

| Resource                                  | Purpose and minimum content                                                                              | Expected output                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Human–AI Decision-Support Specification   | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Dashboard Signal-to-Action Record         | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Delegation, Oversight, and Action Latency | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Human–AI Decision-Support Specification*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Dashboard Signal-to-Action Record*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define the operational decisions, decision windows, and accountable owners before selecting indicators. - Separate observed values, model estimates, forecasts, and recommendations. - Create an indicator dictionary with numerator, denominator, exclusions, latency, and limitations. - Map each signal to a feasible action and at least one balancing measure. - Test forecast value against a standardized non-predictive dashboard and targeted process redesign. - Measure action latency, downstream congestion, workload, and the consequences of recommendations that cannot be implemented.

### *Delegation, Oversight, and Action Latency*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### Key Terms

Table: Key terms for Chapter 3

| Term             | Working definition and decision relevance                                                        |
|------------------|--------------------------------------------------------------------------------------------------|
| Decision support | Information intended to influence a defined decision and action within a meaningful time window. |
| Delegation       | The allocation of tasks and authority between people and technology.                             |
| Action latency   | The elapsed time from signal generation to feasible response.                                    |

### Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 3

| Instrument              | Status or type      | Appropriate use and limitation                                    |
|-------------------------|---------------------|-------------------------------------------------------------------|
| DECIDE-AI               | Reporting guideline | Supports transparent reporting of early live clinical evaluation. |
| Decision-curve analysis | Analytical method   | Examines threshold-specific net benefit.                          |
| Safety case             | Assurance argument  | Links hazards, controls, and evidence for a defined context.      |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                       | Purpose                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-C04 — Human Oversight and Accountability Matrix<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                         | Assign accountability, responsibility, consultation, information, escalation, pause, update, restart, and retirement authority.      |
| AIHE-CASE03 — Deterioration Alert Under Constrained Review Capacity<br>Worked case; md; csv; json; xlsx                        | Which alert threshold produces an operationally safe balance between missed deterioration and review burden?                         |
| AIHE-CASE03B — Hospital Command Dashboard and Patient Flow<br>Worked case; md; csv; json; xlsx                                 | Evaluate whether and under what conditions a command dashboard should be scaled as a permanent patient-flow management intervention. |
| AIHE-O — Human Oversight and Accountability Workbook<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Assign accountable and responsible owners, consultation, information, escalation, pause, update, restart, and retirement authority.  |

| Resource                                                                                                                               | Purpose                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-PY06 — Decision-curve analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                           | Compares prediction-model net benefit with treat-all and treat-none strategies across clinically relevant thresholds.                                            |
| AIHE-PY21 — Discrete-event simulation of hospital flow<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                        | Simulates arrivals, queues, service times, capacity, prioritization, and delays to assess how AI-enabled triage or scheduling affects operations.                |
| AIHE-PY34 — Time-series forecasting for demand, capacity, and budget planning<br>Python analytics; py; ipynb; xlsx; csv; json; png; md | Fits simple trend and exponential-smoothing forecasts to operational or financial time series and compares forecast accuracy.                                    |
| AIHE-PY35 — Constrained resource-allocation optimization<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                      | Selects an allocation across candidate AI or service options to maximize expected value subject to budget, workforce, capacity, and minimum-service constraints. |

## Part II: Organizational Performance, Workforce, and Quality

This Part examines whether proposed capability can be converted into productive work, quality, accreditation evidence, patient safety, and organizational learning.

### Chapter 4: Productivity, Work Redesign, and Implementation

#### Chapter Orientation

Chapter 4 connects productivity claims with workforce redesign and implementation. The resources separate gross time saved from review and correction, usable capacity, capacity actually redeployed, quality-adjusted outcomes, and the implementation work required to sustain benefit.

Use this suite for documentation tools, scheduling systems, administrative automation, and other proposals built around efficiency. It supports task decomposition, workload-transfer analysis, implementation strategy selection, training, adaptation, sustainment, and a defensible benefit-realization plan.

#### Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                      | How to use it                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Ambient Documentation and Workforce Redesign                                          | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Productivity-Realization Chain; Determinant-to-Strategy Matrix; Implementation and Sustainment Record | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | Productivity, Implementation determinant, Sustainment                                                 | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |

| Layer               | Chapter resource                                                                             | How to use it                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Repository coverage | 1 core, 3 python analytics, 1 reference guide, 2 supplemental specialist tool, 1 worked case | Use the catalog map below to locate editable, process-flow, data, and computational variants. |

## Worked Example: Ambient Documentation and Workforce Redesign

### Purpose and Status

An ambulatory network is considering an ambient documentation system to reduce after-hours work. The vendor reports substantial drafting-time savings, while clinicians are concerned about correction burden, privacy, coding changes, and pressure to convert every saved minute into additional appointments.

### Decision Problem and Comparators

Decision problem. Does the intervention create net workforce and organizational value, and what workflow, governance, and implementation conditions are required before scale?

Table: Comparators for the Chapter 4 worked example

| Option | Comparator or strategy                                       |
|--------|--------------------------------------------------------------|
| 1      | Current documentation workflow                               |
| 2      | Template and workflow redesign without generative technology |
| 3      | Additional transcription or administrative support           |
| 4      | Ambient documentation with opt-in human review               |

### Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 4 worked example

| Parameter                              | Base value        | Source or interpretation |
|----------------------------------------|-------------------|--------------------------|
| Clinicians in pilot                    | 60                | Pilot plan               |
| Baseline documentation time per visit  | 11.5 minutes      | Time study               |
| Vendor-reported gross saving           | 5.0 minutes/visit | Vendor evidence          |
| Observed review and correction time    | 2.4 minutes/visit | Local usability test     |
| Monthly subscription per clinician     | EUR 420           | Proposed contract        |
| Average visits per clinician per month | 310               | Scheduling system        |

| Parameter                                | Base value | Source or interpretation |
|------------------------------------------|------------|--------------------------|
| Clinically consequential correction rate | 1.8%       | Pilot quality audit      |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                              | Decision record                               |
|------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Map the full documentation workflow and distinguish drafting assistance from coding or downstream recommendations.         | Evidence and responsible owner to be recorded |
| 2    | Measure gross time, review, correction, escalation, and coordination by professional role.                                 | Evidence and responsible owner to be recorded |
| 3    | Assess whether net time can be pooled or scheduled as usable capacity.                                                     | Evidence and responsible owner to be recorded |
| 4    | Evaluate quality, privacy, consent, patient interaction, and after-hours work alongside throughput.                        | Evidence and responsible owner to be recorded |
| 5    | Use CFIR, NASSS-CAT, or another selected implementation resource for a defined purpose rather than as a generic checklist. | Evidence and responsible owner to be recorded |
| 6    | Define benefit owners, renewal thresholds, and controls for model, prompt, and template updates.                           | Evidence and responsible owner to be recorded |

### Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                                   | Result            | Interpretation                              |
|------------------------------------------|-------------------|---------------------------------------------|
| Gross time saved                         | 5.0 minutes/visit | Vendor claim                                |
| Net time saved after review              | 2.6 minutes/visit | Local pilot estimate                        |
| Potential monthly net time per clinician | 13.4 hours        | Before accounting for downtime and meetings |

| Output                        | Result                                                    | Interpretation                    |
|-------------------------------|-----------------------------------------------------------|-----------------------------------|
| Cash-releasing saving         | Not established                                           | No payroll reduction demonstrated |
| Main realized value candidate | Reduced after-hours work and improved patient interaction | Requires longitudinal evidence    |

### *Interpretation and Recommendation*

Continue a limited opt-in pilot and do not treat gross time saving as a cash benefit. Scale only if net time benefit, documentation quality, patient acceptability, privacy, and workforce experience remain favourable and no adverse coding or downstream-utilization signal emerges.

- documentation assistance remains separate from coding and order automation.
- clinically consequential content requires meaningful professional confirmation.
- specialty-specific correction patterns and subgroup effects are monitored.
- released time has a documented and acceptable use.
- renewal is linked to sustained net benefit after the novelty period.

### **Supplemental Resources**

These supplemental resources translate the Chapter 4 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 4 supplemental-resource set

| Resource                              | Purpose and minimum content                                                                              | Expected output                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Productivity-Realization Chain        | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Determinant-to-Strategy Matrix        | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Implementation and Sustainment Record | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Productivity-Realization Chain*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Determinant-to-Strategy Matrix*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Map the full documentation workflow and distinguish drafting assistance from coding or downstream recommendations. - Measure gross time, review, correction, escalation, and coordination by professional role. - Assess whether net time

can be pooled or scheduled as usable capacity. - Evaluate quality, privacy, consent, patient interaction, and after-hours work alongside throughput. - Use CFIR, NASSS-CAT, or another selected implementation resource for a defined purpose rather than as a generic checklist. - Define benefit owners, renewal thresholds, and controls for model, prompt, and template updates.

### *Implementation and Sustainment Record*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### *Key Terms*

Table: Key terms for Chapter 4

| Term                       | Working definition and decision relevance                                                   |
|----------------------------|---------------------------------------------------------------------------------------------|
| Productivity               | Output or value generated relative to resources, adjusted for quality and transferred work. |
| Implementation determinant | A factor that impedes or enables implementation in a defined context.                       |
| Sustainment                | Continued appropriate use and benefit after initial implementation support declines.        |

### *Status and Appropriate Use of Selected Instruments*

Table: Selected instruments relevant to Chapter 4

| Instrument   | Status or type                                | Appropriate use and limitation                                        |
|--------------|-----------------------------------------------|-----------------------------------------------------------------------|
| MRC guidance | Methodological guidance                       | Supports development and evaluation of complex interventions.         |
| CFIR         | Determinant framework                         | Identifies barriers and facilitators.                                 |
| NASSS-CAT    | Complexity assessment tool                    | Examines nonadoption, abandonment, scale, spread, and sustainability. |
| RE-AIM/PRISM | Implementation-outcome and context frameworks | Support real-world implementation and maintenance evaluation.         |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                 | Purpose                                                                                                                |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| AIHE-B03 — Benefit Realization and Capacity Conversion<br>Core; docx; xlsx; csv; json; md; mmd; svg; png | Translate claimed benefits into measurable mechanisms, owners, baselines, conversion conditions, and realized results. |

| Resource                                                                                                                              | Purpose                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-CASE04 — Ambient Documentation Assistant: Productivity and Work Redesign<br>Worked case; md; csv; json; xlsx                     | How much of the claimed notes time saving becomes usable capacity after review, correction, adoption, and implementation costs?                     |
| AIHE-RG03 — Implementation Framework Selection Guide<br>Reference guide; docx; xlsx; csv; json; md                                    | Select CFIR, NASSS-CAT, RE-AIM/PRISM, TAM/UTAUT, and mixed-method approaches according to the implementation question rather than name recognition. |
| AIHE-EX07 — Benefit Realization and Capacity Conversion Log<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Assign each benefit to a mechanism, measure, owner, baseline, timing, accounting treatment, and verification source.                                |
| AIHE-Z — ROI and Workflow Value Worksheet<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                   | Distinguish gross time saved, review burden, capacity released, capacity used, and realized value.                                                  |
| AIHE-PY22 — System-dynamics model of adoption, capacity, and value<br>Python analytics; py; ipynb; xlsx; csv; json; png; md           | Explores feedback among training, trust, adoption, workforce capacity, workload, benefit realization, and support burden over time.                 |
| AIHE-PY23 — Agent-based simulation of clinician adoption<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                     | Models heterogeneous clinicians whose adoption changes through training, peer influence, perceived usefulness, workload, and adverse incidents.     |
| AIHE-PY28 — Workflow time, capacity, productivity, and benefit realization<br>Python analytics; py; ipynb; xlsx; csv; json; png; md   | Translates gross task-time savings into net time, usable capacity, realized output, quality-adjusted productivity, and economic value.              |

## Chapter 5: Quality Management, Accreditation, and Organizational Learning

### Chapter Orientation

Chapter 5 places AI-enabled services within quality management, accreditation, patient safety, corrective action, audit, and organizational learning. The resources link technical and clinical indicators with workflow, equity, patient experience, economic value, incident investigation, and verified corrective action.

Use this suite after a pilot or deployment, after a model or workflow update, or when monitoring identifies an unexpected signal. It helps distinguish statistical variation from unacceptable performance and connects investigation, containment, corrective action, revalidation, communication, and closure.

### Resource Suite at a Glance

| Layer          | Chapter resource                                                 | How to use it                                                                                                  |
|----------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example | Worked Example: Quality Signal After a Model and Workflow Update | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |

| Layer                  | Chapter resource                                                                                    | How to use it                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Supplemental resources | Quality Monitoring Plan; Incident and CAPA Record; Accreditation and Organizational-Learning Record | Create the institutional records needed for analysis, approval, monitoring, and review.       |
| Reference focus        | Quality assurance, Quality control, Organizational learning                                         | Clarify terminology, method selection, instrument status, formulas, and update requirements.  |
| Repository coverage    | 2 core, 1 python analytics, 1 supplemental specialist tool, 1 worked case                           | Use the catalog map below to locate editable, process-flow, data, and computational variants. |

### Worked Example: Quality Signal After a Model and Workflow Update

#### *Purpose and Status*

A deterioration-alert service has been used for eighteen months. Following an electronic health-record upgrade and a model update, the false-alert rate rises and nurses report more workarounds, while aggregate discrimination appears unchanged.

#### *Decision Problem and Comparators*

Decision problem. Should the service continue unchanged, be restricted, undergo corrective action and revalidation, or be suspended pending investigation?

Table: Comparators for the Chapter 5 worked example

| Option | Comparator or strategy                                    |
|--------|-----------------------------------------------------------|
| 1      | Continue current operation                                |
| 2      | Revert to the previous model and interface version        |
| 3      | Restrict use to selected units                            |
| 4      | Suspend and return to the conventional escalation pathway |

#### *Synthetic Parameters and Evidence Sources*

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 5 worked example

| Parameter                    | Base value | Source or interpretation    |
|------------------------------|------------|-----------------------------|
| Baseline false-alert rate    | 14%        | Quality dashboard           |
| Post-update false-alert rate | 23%        | Four-week monitoring window |

| Parameter                         | Base value      | Source or interpretation |
|-----------------------------------|-----------------|--------------------------|
| Aggregate AUROC before/after      | 0.82 / 0.81     | Monitoring report        |
| Median response time before/after | 12 / 19 minutes | Workflow log             |
| Units reporting workarounds       | 5 of 8          | Safety walk-rounds       |
| Serious incidents                 | 0               | Incident register        |
| Near-miss reports                 | 7               | Patient-safety system    |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                                                        | Decision record                               |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Reconstruct the change timeline, data sources, model, software, interface, threshold, and workflow versions.                                         | Evidence and responsible owner to be recorded |
| 2    | Distinguish statistical stability from clinical and managerial acceptability.                                                                        | Evidence and responsible owner to be recorded |
| 3    | Investigate source-data, interface, threshold, user, staffing, and downstream-capacity explanations before selecting corrective action.              | Evidence and responsible owner to be recorded |
| 4    | Use the existing quality and patient-safety system to document containment, root causes, corrective and preventive action, and effectiveness checks. | Evidence and responsible owner to be recorded |
| 5    | Assess whether the change is material and which validation evidence must be repeated.                                                                | Evidence and responsible owner to be recorded |
| 6    | Communicate the decision, residual uncertainty, and conditions for return to unrestricted use.                                                       | Evidence and responsible owner to be recorded |

### Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                   | Result                                   | Interpretation                                    |
|--------------------------|------------------------------------------|---------------------------------------------------|
| Technical discrimination | Broadly stable                           | Does not rule out workflow or calibration failure |
| Operational performance  | Deteriorated                             | Response delay and alert burden increased         |
| Likely cause             | Combined interface and threshold effects | Requires confirmation                             |
| Immediate action         | Restrict and revert where feasible       | Containment while investigation proceeds          |
| Quality decision         | CAPA and targeted revalidation           | Not automatic retraining                          |

### *Interpretation and Recommendation*

Restrict use, restore the previous validated configuration where safe, and complete a formal incident and corrective-action process. Unrestricted use should resume only after the source of the signal is understood and the effectiveness of corrective action is verified.

- versioned reconstruction of data, software, model, interface, and workflow.
- unit-level and subgroup analysis.
- documented CAPA with named owners and completion dates.
- verification that response time and alert burden return within acceptable limits.
- executive authority to suspend or retire the intervention if performance remains unacceptable.

### **Supplemental Resources**

These supplemental resources translate the Chapter 5 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 5 supplemental-resource set

| Resource                                         | Purpose and minimum content                                                                              | Expected output                                       |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Quality Monitoring Plan                          | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Incident and CAPA Record                         | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Accreditation and Organizational-Learning Record | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### Quality Monitoring Plan

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### Incident and CAPA Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Reconstruct the change timeline, data sources, model, software, interface, threshold, and workflow versions. - Distinguish statistical stability from clinical and managerial acceptability. - Investigate source-data, interface, threshold, user, staffing, and downstream-capacity explanations before selecting corrective action. - Use the existing quality and patient-safety system to document containment, root causes, corrective and preventive action, and effectiveness checks. - Assess whether the change is material and which validation evidence must be repeated. - Communicate the decision, residual uncertainty, and conditions for return to unrestricted use.

### Accreditation and Organizational-Learning Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### Key Terms

Table: Key terms for Chapter 5

| Term                    | Working definition and decision relevance                                                   |
|-------------------------|---------------------------------------------------------------------------------------------|
| Quality assurance       | Evidence supporting confidence that requirements are being met.                             |
| Quality control         | Operational detection of variation or failure against statistical and acceptability limits. |
| Organizational learning | Conversion of monitoring and experience into revised knowledge, practice, and governance.   |

### Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 5

| Instrument    | Status or type                                  | Appropriate use and limitation                                              |
|---------------|-------------------------------------------------|-----------------------------------------------------------------------------|
| ISO 7101      | Voluntary healthcare management-system standard | Supports quality-management requirements.                                   |
| ISO/IEC 42001 | Voluntary AI management-system standard         | Supports organization-wide AI management responsibilities.                  |
| SPC           | Statistical method                              | Supports detection of process variation; not a clinical acceptability rule. |
| CAPA          | Quality process                                 | Links investigation to corrective and preventive action and verification.   |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                              | Purpose                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-C01 — AI Quality-Control and Accreditation Evidence Plan<br>Core; docx; xlsx; csv; json; md; mmd; svg; png       | Integrate AI-enabled services into quality control, accreditation evidence, monitoring, thresholds, action, and closure.                                          |
| AIHE-C02 — Corrective and Preventive Action Record<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                  | Document nonconformity, containment, causal analysis, corrective and preventive action, effectiveness, residual risk, and closure.                                |
| AIHE-CASE05 — AI Quality Management, Accreditation, and Corrective Action<br>Worked case; md; csv; json; xlsx         | When should a change in performance trigger investigation, restriction, and corrective action within the quality-management system?                               |
| AIHE-I — Health-System AI Indicator Library<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Select outcome, process, balancing, equity, workforce, economic, governance, and lifecycle indicators with definitions, owners, sources, thresholds, and actions. |
| AIHE-PY02 — Descriptive statistics and visual comparisons<br>Python analytics; py; ipynb; xlsx; csv; json; png; md    | Summarizes numerical and categorical variables, compares groups, and creates baseline visual diagnostics before modeling.                                         |

## Part III: Economic Evaluation, Decision Modelling, and Procurement

This Part develops lifecycle costing, economic evaluation, modelling under uncertainty, procurement, contracting, and vendor governance.

### Chapter 6: Lifecycle Cost, Pricing, and Benefit Realization

#### Chapter Orientation

Chapter 6 develops the full lifecycle cost and financial case. The resources identify costs that sit outside the licence price, model volume and pricing exposure, distinguish cash savings from avoided cost and released capacity, and assign responsibility for converting projected benefit into observed value.

Use this suite during business-case development, pricing negotiation, pilot budgeting, renewal, or exit. The records should preserve the perspective, price year, time horizon, cost boundary, allocation rules, volume scenarios, benefit mechanism, budget owner, uncertainty, and residual value.

#### Resource Suite at a Glance

| Layer          | Chapter resource                                                                             | How to use it                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example | Worked Examples: Clinical Documentation Platform; Financial Recognition and Budget Incidence | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |

| Layer                  | Chapter resource                                                               | How to use it                                                                                 |
|------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Supplemental resources | Lifecycle Cost Log; Pricing and Volume Scenarios; Benefit-Realization Register | Create the institutional records needed for analysis, approval, monitoring, and review.       |
| Reference focus        | Lifecycle cost, Pricing exposure, Benefit realization                          | Clarify terminology, method selection, instrument status, formulas, and update requirements.  |
| Repository coverage    | 2 core, 2 Python analytics, 3 supplemental specialist tools, 2 worked cases    | Use the catalog map below to locate editable, process-flow, data, and computational variants. |

### Worked Example: Clinical Documentation Platform: Lifecycle Cost and Pricing

#### Purpose and Status

A hospital group is comparing three pricing proposals for a documentation platform: a fixed enterprise licence, a per-clinician subscription, and a hybrid base fee plus per-encounter charge. Each proposal excludes some implementation, monitoring, and exit activities.

#### Decision Problem and Comparators

Decision problem. Which pricing and contracting structure produces the most predictable and defensible lifecycle cost, and under what conditions could projected benefits be realized?

Table: Comparators for the Chapter 6 worked example

| Option | Comparator or strategy                                   |
|--------|----------------------------------------------------------|
| 1      | Fixed enterprise licence                                 |
| 2      | Per-clinician subscription                               |
| 3      | Hybrid base plus encounter-based fee                     |
| 4      | Do not procure; invest in workflow and template redesign |

#### Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 6 worked example

| Parameter                  | Base value            | Source or interpretation |
|----------------------------|-----------------------|--------------------------|
| Clinicians                 | 420                   | Workforce plan           |
| Annual eligible encounters | 520,000               | Activity forecast        |
| Enterprise base fee        | EUR 1.15 million/year | Vendor A                 |
| Per-clinician fee          | EUR 390/month         | Vendor B                 |

| Parameter                          | Base value         | Source or interpretation |
|------------------------------------|--------------------|--------------------------|
| Hybrid base fee                    | EUR 540,000/year   | Vendor C                 |
| Hybrid variable fee                | EUR 1.15/encounter | Vendor C                 |
| Initial integration and governance | EUR 760,000        | Local estimate           |
| Annual monitoring and support      | EUR 310,000        | Local estimate           |
| Expected useful decision horizon   | 5 years            | Business case            |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                                                | Decision record                               |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Define perspective, price year, time horizon, comparator, and full cost boundary.                                                            | Evidence and responsible owner to be recorded |
| 2    | Classify costs as fixed, step-fixed, variable, contingent, shared, or exit related.                                                          | Evidence and responsible owner to be recorded |
| 3    | Model low, expected, and high activity for each pricing structure.                                                                           | Evidence and responsible owner to be recorded |
| 4    | Discount costs and credible monetised benefits; preserve non-monetised value separately.                                                     | Evidence and responsible owner to be recorded |
| 5    | Create a benefit-realization register that distinguishes cash savings, avoided costs, usable capacity, quality, workforce, and public value. | Evidence and responsible owner to be recorded |
| 6    | Test price, uptake, review burden, implementation delay, support, and exit thresholds.                                                       | Evidence and responsible owner to be recorded |

### Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                            | Result                                            | Interpretation                                      |
|-----------------------------------|---------------------------------------------------|-----------------------------------------------------|
| Most predictable annual spend     | Enterprise licence                                | May overpay at low uptake                           |
| Lowest expected five-year PV cost | Hybrid proposal                                   | Dependent on activity cap and included support      |
| Dominant uncertainty              | Net usable capacity per encounter                 | Not vendor price alone                              |
| Cash-releasing saving             | Not established                                   | Released time is not yet linked to budget reduction |
| Preferred decision                | Negotiate hybrid with cap and conditional renewal | Preserves predictability and proportionality        |

### *Interpretation and Recommendation*

Negotiate a capped hybrid structure with explicit integration, monitoring, updates, data access, and exit support. Treat productivity benefit as conditional until local evidence shows how net time is converted into usable capacity or another measurable outcome.

- all pricing units are clinically and financially understandable.
- base and variable components do not double charge essential services.
- low, expected, and high activity scenarios remain affordable.
- benefit owners and verification sources are named.
- renewal depends on observed value, current evidence, and acceptable exit options.

### **Supplemental Resources**

These supplemental resources translate the Chapter 6 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 6 supplemental-resource set

| Resource                     | Purpose and minimum content                                                                              | Expected output                                       |
|------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Lifecycle Cost Log           | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Pricing and Volume Scenarios | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Benefit-Realization Register | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Lifecycle Cost Log*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Pricing and Volume Scenarios*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define perspective, price year, time horizon, comparator, and full cost boundary. - Classify costs as fixed, step-fixed, variable, contingent, shared, or exit related. - Model low, expected, and high activity for each pricing structure. - Discount costs and credible monetised benefits; preserve non-monetised value separately. - Create a benefit-realization register that distinguishes cash savings, avoided costs, usable capacity, quality, workforce, and public value. - Test price, uptake, review burden, implementation delay, support, and exit thresholds.

### *Benefit-Realization Register*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## **Reference Resources**

### *Key Terms*

Table: Key terms for Chapter 6

| Term                | Working definition and decision relevance                                               |
|---------------------|-----------------------------------------------------------------------------------------|
| Lifecycle cost      | Incremental resources from preparation through operation, change, transition, and exit. |
| Pricing exposure    | How contractual units and activity translate into financial risk.                       |
| Benefit realization | Evidence that a projected mechanism produced an observable and usable result.           |

### *Status and Appropriate Use of Selected Instruments*

Table: Selected instruments relevant to Chapter 6

| Instrument | Status or type                     | Appropriate use and limitation                                            |
|------------|------------------------------------|---------------------------------------------------------------------------|
| TCAIO      | Author-developed costing framework | Organizes full lifecycle cost from acquisition through exit.              |
| CHEERS-AI  | Reporting guideline                | Supports transparent reporting of AI economic evaluations.                |
| TDABC      | Costing method                     | Estimates personnel and activity costs from time and capacity cost rates. |

## **Repository Resource Map**

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                      | Purpose                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-B01 — Economic Evaluation Specification<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                                | Define the economic decision, perspective, comparator, horizon, outcomes, costs, methods, uncertainty, safeguards, and validation.                                |
| AIHE-B02 — Lifecycle Cost and Pricing Record<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                                | Record acquisition through exit costs, resource use, pricing exposure, opportunity cost, residual value, and evidence.                                            |
| AIHE-CASE06 — Five-Year Lifecycle Cost, Pricing, ROI, and Benefit Realization<br>Worked case; md; csv; json; xlsx             | Does the proposed AI service create value after all lifecycle costs, pricing exposure, and benefit-conversion conditions are included?                            |
| AIHE-CASE06B — Financial Recognition and Budget Incidence<br>Worked case; md; csv; json; xlsx                                 | How should released capacity, avoided resource use, cashable savings, revenue, accounting treatment, and budget incidence be distinguished?                       |
| AIHE-B — Economic Evaluation Specification<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png          | Specify the decision, perspective, comparator, costs, outcomes, horizon, uncertainty, distribution, and reporting plan.                                           |
| AIHE-EX17 — AI Pricing and Contract Scenario Record<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Model base and variable fees in clinically legible units, cap exposure, price support explicitly, and link renewal to realized value.                             |
| AIHE-M — TCAIO Costing Workbook<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                     | Estimate incremental lifecycle costs from acquisition through operation, monitoring, material change, and exit.                                                   |
| AIHE-PY10 — Total Cost of AI Ownership (TCAIO) lifecycle costing<br>Python analytics; py; ipynb; xlsx; csv; json; png; md     | Calculates one-time and recurring AI costs, discounted present value, cost per use, and optimistic/base/conservative scenarios across the full lifecycle.         |
| AIHE-PY11 — ROI, net benefit, and benefit-realization analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md       | Calculates gross and risk-adjusted benefits, total cost, net benefit, ROI, payback period, and the effect of realization probability and attribution assumptions. |

## Chapter 7: Economic Evaluation, Affordability, and Health Technology Assessment

### Chapter Orientation

Chapter 7 separates value for money, budget impact, affordability, health technology assessment, reimbursement, and local adoption. The resources support a transparent economic-evaluation protocol and ensure that cost-effectiveness does not obscure cash-flow constraints, implementation capacity, distributional effects, or unresolved safeguards.

Use this suite when an institution must decide whether to adopt, pilot, defer, redesign, restrict, or reject an intervention. The outputs should show incremental costs and outcomes, annual budget consequences, evidence maturity, uncertainty, subgroup effects, payment context, and local prerequisites.

## Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                        | How to use it                                                                                                  |
|------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Remote Monitoring: Economic Evaluation, Affordability, and Health Technology Assessment | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Economic Evaluation Protocol; Budget-Impact Record; HTA and Local-Adoption Decision Record              | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | Cost-effectiveness, Affordability, HTA                                                                  | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |
| Repository coverage    | 1 core, 3 python analytics, 2 supplemental specialist tool, 1 worked case                               | Use the catalog map below to locate editable, process-flow, data, and computational variants.                  |

## Worked Example: Remote Monitoring: Economic Evaluation, Affordability, and Health Technology Assessment

### Purpose and Status

A public payer and hospital network are assessing an AI-supported remote-monitoring programme for people with heart failure. The programme may reduce emergency visits and admissions, but it requires devices, connectivity, monitoring staff, escalation capacity, and recurring platform fees.

### Decision Problem and Comparators

Decision problem. Is the programme cost-effective, affordable, and suitable for local adoption, and which evidence gaps justify a staged decision?

Table: Comparators for the Chapter 7 worked example

| Option | Comparator or strategy                               |
|--------|------------------------------------------------------|
| 1      | Usual scheduled follow-up                            |
| 2      | Non-AI remote monitoring with threshold rules        |
| 3      | AI-supported remote monitoring with nurse review     |
| 4      | Enhanced community nursing without remote monitoring |

### Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 7 worked example

| Parameter                        | Base value        | Source or interpretation     |
|----------------------------------|-------------------|------------------------------|
| Eligible population              | 8,000             | Commissioning estimate       |
| Year-1 uptake                    | 35%               | Implementation scenario      |
| Incremental cost per participant | EUR 1,180/year    | Programme costing            |
| Expected admissions avoided      | 0.10/person-year  | Evidence synthesis           |
| QALY gain                        | 0.028/person-year | Economic model               |
| Cost per admission               | EUR 5,600         | Payer data                   |
| Implementation cost              | EUR 1.4 million   | Network estimate             |
| Decision threshold               | EUR 35,000/QALY   | Illustrative local threshold |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                               | Decision record                               |
|------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Specify decision-maker, perspective, comparator, population, horizon, and intended scale.                                   | Evidence and responsible owner to be recorded |
| 2    | Estimate incremental costs, effects, QALYs, and net benefit.                                                                | Evidence and responsible owner to be recorded |
| 3    | Present annual and cumulative budget impact under uptake scenarios.                                                         | Evidence and responsible owner to be recorded |
| 4    | Separate economic cost from cash flow and productive capacity from cash-releasing savings.                                  | Evidence and responsible owner to be recorded |
| 5    | Assess subgroup access, connectivity, digital support, and distribution of opportunity cost.                                | Evidence and responsible owner to be recorded |
| 6    | Identify whether remaining uncertainty should lead to adoption, a restricted pilot, price negotiation, or further research. | Evidence and responsible owner to be recorded |

### Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                                    | Result                                                   | Interpretation                                     |
|-------------------------------------------|----------------------------------------------------------|----------------------------------------------------|
| Incremental programme cost before offsets | EUR 9.44 million/year at full scale                      | Excludes initial implementation                    |
| Expected admission offset                 | EUR 4.48 million/year at full scale                      | Dependent on causal effectiveness                  |
| Illustrative ICER                         | Approximately EUR 22,900/QALY                            | Favourable at stated threshold                     |
| Year-1 budget impact                      | Positive and substantial                                 | May remain unaffordable despite cost-effectiveness |
| Preferred decision                        | Phased adoption with negotiated price and access support | HTA and affordability separated                    |

### *Interpretation and Recommendation*

Support phased adoption where the comparative clinical evidence is credible, but link scale to annual affordability, observed service capacity, subgroup access, and updated budget impact. The payer and provider should align payment and implementation responsibilities.

- the exact intervention bundle and comparator are defined.
- annual and cumulative budget impact remain within an agreed envelope.
- non-digital access and connectivity support are funded.
- local causal and implementation evidence is collected.
- price, uptake, and service-capacity thresholds are included in the decision record.

### **Supplemental Resources**

These supplemental resources translate the Chapter 7 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 7 supplemental-resource set

| Resource                               | Purpose and minimum content                                                                              | Expected output                                       |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Economic Evaluation Protocol           | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Budget-Impact Record                   | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| HTA and Local-Adoption Decision Record | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

## Economic Evaluation Protocol

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### Budget-Impact Record

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Specify decision-maker, perspective, comparator, population, horizon, and intended scale. - Estimate incremental costs, effects, QALYs, and net benefit. - Present annual and cumulative budget impact under uptake scenarios. - Separate economic cost from cash flow and productive capacity from cash-releasing savings. - Assess subgroup access, connectivity, digital support, and distribution of opportunity cost. - Identify whether remaining uncertainty should lead to adoption, a restricted pilot, price negotiation, or further research.

### HTA and Local-Adoption Decision Record

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### Key Terms

Table: Key terms for Chapter 7

| Term               | Working definition and decision relevance                                                |
|--------------------|------------------------------------------------------------------------------------------|
| Cost-effectiveness | Comparison of incremental outcomes and costs relative to an alternative.                 |
| Affordability      | Whether the responsible budget and implementation capacity can support the intervention. |
| HTA                | Multidisciplinary appraisal supporting policy, reimbursement, and adoption decisions.    |

### Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 7

| Instrument          | Status or type                         | Appropriate use and limitation                                                  |
|---------------------|----------------------------------------|---------------------------------------------------------------------------------|
| CHEERS 2022         | Reporting guideline                    | Supports transparent reporting of health-economic evaluations.                  |
| CHEERS-AI           | AI-specific reporting extension        | Adds AI intervention, implementation, updating, and uncertainty considerations. |
| NICE reference case | Jurisdiction-specific methods guidance | Supports consistent appraisal in England; not universal.                        |
| EU HTA Regulation   | Binding EU regulation in scope         | Supports joint clinical assessment while national decisions remain.             |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                                 | Purpose                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-B04 — Cost-Effectiveness, Budget Impact, and HTA Record<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                           | Summarize cost-effectiveness, affordability, distribution, HTA, reimbursement, adoption conditions, uncertainty, and recommendation.                                                                        |
| AIHE-CASE07 — Remote Monitoring: Cost-Effectiveness, Budget Impact, and Health Technology Assessment<br>Worked case; md; csv; json; xlsx | Can an intervention be cost-effective yet remain unaffordable or unsuitable for local adoption?                                                                                                             |
| AIHE-AA — Cost-Effectiveness and Budget-Impact Tool<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png            | Compare incremental costs and outcomes, net benefit, affordability, uptake, cash flow, and sensitivity scenarios.                                                                                           |
| AIHE-EX18 — HTA, Reimbursement, and Adoption Record<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png            | Distinguish regulatory authorization, HTA, reimbursement or payment, procurement, and local adoption; record the authority, perspective, evidence, gaps, and continuing evidence conditions for each layer. |
| AIHE-PY12 — Cost-effectiveness, ICER, and net monetary benefit<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                  | Compares an AI-enabled pathway with a comparator using incremental cost, incremental effect, ICER, net monetary benefit, and dominance rules.                                                               |
| AIHE-PY13 — Multiyear budget-impact analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                    | Estimates annual and cumulative financial consequences for a specific budget holder under low, base, and high adoption or effectiveness scenarios.                                                          |
| AIHE-PY16 — Distributional cost-effectiveness and equity-weighted analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md      | Compares costs, effects, net monetary benefit, access, and equity-weighted value across population groups.                                                                                                  |

## Chapter 8: Decision Modelling, Simulation, and the Value of Information

### Chapter Orientation

Chapter 8 explains how the structure of the decision determines the modelling method. The resources cover decision trees, state-transition models, microsimulation, discrete-event simulation, queueing, hybrid models, scenario analysis, probabilistic sensitivity analysis, and value of information.

The detailed imaging-prioritization case is retained as the principal worked example because it shows how queue effects, classification errors, time to diagnosis, capacity, lifetime outcomes, subgroup consequences, and research priorities can be linked without treating image-level performance as the whole decision.

## Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                                | How to use it                                                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Examples: Imaging Prioritization; EVSI and Expected Net Benefit of Sampling                              | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Model Specification; Verification, Validation, and Reproducibility; Uncertainty and Value-of-Information Record | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | Decision model, Simulation, Value of information                                                                | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |
| Repository coverage    | 1 core, 5 Python analytics, 1 reference guide, 5 supplemental specialist tools, 2 worked cases                  | Use the catalog map below to locate editable, process-flow, data, and computational variants.                  |

## Worked Example: Imaging Prioritization

### Purpose and Status

This worked example shows how a health system can combine queue modelling, diagnostic consequences, economic evaluation, subgroup analysis, and decision uncertainty when considering an artificial intelligence (AI) system that prioritises imaging studies for human interpretation. The intervention changes the order in which studies are read; it does not replace professional review or issue a final diagnosis.

All values are synthetic and are included solely to demonstrate the analytical sequence. They are not clinical estimates, price benchmarks, or recommendations for a particular imaging service. Local use would require current clinical evidence, local workflow and capacity data, subgroup validation, approved cost sources, and governance review. The recent United Kingdom breast-screening evaluation cited in Chapter 8 illustrates the value of connecting operational workflow with longer-term health and economic consequences, but the numerical values below are not taken from that study (1).

| Field               | Specification                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Resource identifier | AIHE-CASE08                                                                                               |
| Aligned chapter     | Chapter 8: Decision Modelling, Simulation, and the Value of Information                                   |
| Decision setting    | Annual cohort of screening-imaging examinations                                                           |
| Perspective         | Hospital and publicly funded health-system perspective                                                    |
| Price year          | 2026 pounds sterling (GBP)                                                                                |
| Analytical horizon  | One year of service operation, with lifetime consequences assigned to materially delayed cancer diagnoses |

| Field        | Specification                                                                                        |
|--------------|------------------------------------------------------------------------------------------------------|
| Comparator   | Current first-in, first-out worklist with manual escalation when staff identify urgency              |
| Intervention | AI-supported prioritization followed by unchanged human interpretation and diagnostic responsibility |
| Status       | Synthetic worked example; not a validated institutional model                                        |

### Decision Problem and Comparator

A regional imaging service processes approximately 25,000 screening studies each year. The organization is considering an AI prioritization system that assigns a priority flag when an examination appears more likely to require rapid review. Flagged studies move to the front of the human-reading worklist. All studies are still interpreted by qualified readers, and the intervention does not alter the number of readers, scanners, diagnostic appointments, or treatment slots.

The decision is whether to retain the current worklist, conduct a controlled pilot, or adopt the prioritization system at scale. The analysis must determine whether earlier review of time-sensitive cases produces sufficient health and resource value to justify licensing, integration, monitoring, training, and the opportunity cost imposed on non-prioritised studies.

| Decision element     | Worked specification                                                                                                       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Population           | Adults receiving screening imaging during one service year                                                                 |
| Service problem      | Clinically important studies may remain in the routine queue long enough to reduce the benefit of early diagnosis          |
| Comparator           | First-in, first-out worklist with existing manual escalation                                                               |
| Intervention         | AI priority flag, priority queue, human interpretation, audit, monitoring, and rollback capability                         |
| Excluded uses        | Autonomous diagnosis, removal of human review, automated treatment decisions, or use outside the validated imaging pathway |
| Primary outcome      | Quality-adjusted life years (QALYs) gained by reducing materially delayed diagnoses                                        |
| Operational outcomes | Time to report, priority volume, routine-worklist delay, capacity use, and review burden                                   |
| Economic outcomes    | Incremental cost, incremental QALYs, incremental cost-effectiveness ratio, net monetary benefit, and budget exposure       |
| Equity question      | Whether performance and reporting time differ for groups with more complex studies or poorer access to specialist services |

| Decision element | Worked specification                                                                                       |
|------------------|------------------------------------------------------------------------------------------------------------|
| Decision options | Do not adopt; redesign; conduct a controlled pilot; adopt with conditions; scale; restrict; or discontinue |

## Model Structure

The worked example uses a simplified hybrid structure. Each component answers a different part of the decision problem: - A discrete-event simulation (DES) represents arrivals, daily reading capacity, temporary capacity loss, queue order, and time to report. DES is appropriate because prioritization changes the ordering of events and the use of constrained resources (2). - A short decision tree classifies examinations as true positive, false negative, false positive, or true negative relative to the priority flag. These categories determine which studies receive earlier review and which studies consume priority capacity without being time-sensitive. - A simplified long-term consequence module links reporting delay to the probability of a materially delayed diagnosis. Each material delay is assigned an expected QALY loss and downstream treatment cost. In a full evaluation, this scalar consequence would normally be replaced by a disease-specific state-transition or patient-level model. - Deterministic sensitivity analysis, probabilistic sensitivity analysis (PSA), subgroup analysis, threshold analysis, and expected value of perfect information (EVPI) examine whether uncertainty could change the decision.

| Component          | What it represents                                                     | Principal output                                                                                      |
|--------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Queue simulation   | Arrival variability, reading capacity, downtime, and priority ordering | Time-to-report distributions for all studies and for priority, routine, cancer, and non-cancer groups |
| Decision tree      | Priority-flag performance and immediate consequences                   | Expected true-positive, false-negative, false-positive, and true-negative counts                      |
| Consequence module | Relationship between delay and lifetime consequences                   | Material delays, QALY loss avoided, and downstream costs avoided                                      |
| Economic module    | Programme costs, offsets, and health outcomes                          | Incremental cost, ICER, and net monetary benefit                                                      |
| Uncertainty module | Parameter and structural uncertainty                                   | Sensitivity results, probability of cost-effectiveness, EVPI, and research priorities                 |

## Parameters and Assumptions

The following table records the base-case values and the ranges used in deterministic or probabilistic analysis. The values are deliberately transparent so that readers can reproduce the calculations and identify which assumptions should be replaced by local evidence.

Table: Synthetic parameter table

| Parameter           | Base value | Sensitivity or PSA specification | Role in the model                        |
|---------------------|------------|----------------------------------|------------------------------------------|
| Annual examinations | 25,000     | Fixed in base PSA                | Service volume                           |
| Working days        | 250        | Fixed                            | Converts annual volume to daily arrivals |

| Parameter                             | Base value  | Sensitivity or PSA specification                        | Role in the model                                |
|---------------------------------------|-------------|---------------------------------------------------------|--------------------------------------------------|
| Cancer prevalence                     | 0.008       | Beta(80, 9920)                                          | Number of time-sensitive cases                   |
| Priority sensitivity                  | 0.90        | Beta(180, 20); DSA 0.80–0.95                            | Probability that a cancer case receives priority |
| Priority specificity                  | 0.92        | Beta(920, 80); DSA 0.88–0.96                            | Controls false-priority volume                   |
| Mean daily arrivals                   | 100         | Poisson arrivals                                        | Queue demand                                     |
| Mean daily reading capacity           | 101.4       | Normal mean 101.4, SD 7, truncated at zero              | Queue service capacity                           |
| Capacity-loss days                    | 5%          | 25% capacity reduction on affected days                 | Operational disruption                           |
| Baseline mean cancer report time      | 5.0 days    | Normal mean 5.0, SD 1.0, minimum 0.5                    | Delay under the comparator                       |
| True-positive report time             | 0.5 day     | Normal mean 0.5, SD 0.15, minimum 0.2                   | Delay for correctly prioritised cancer cases     |
| False-negative report time            | 5.44 days   | Normal mean 5.44, SD 1.0, minimum 0.5                   | Delay for missed priority cases                  |
| QALY loss per material delay          | 0.10        | Normal mean 0.10, SD 0.03, minimum 0.01; DSA 0.05–0.15  | Lifetime health consequence                      |
| Cost per material delay               | GBP 2,500   | Normal mean 2,500, SD 750, minimum 250; DSA 1,000–5,000 | Downstream treatment consequence                 |
| Annual licence                        | GBP 65,000  | Triangular(40,000; 65,000; 90,000)                      | Recurring programme cost                         |
| Annualised integration                | GBP 20,000  | Normal mean 20,000, SD 4,000, minimum 5,000             | Interface and technical support                  |
| Monitoring and assurance              | GBP 18,000  | Normal mean 18,000, SD 3,600, minimum 5,000             | Performance, equity, safety, and incident review |
| Training and service support          | GBP 8,000   | Normal mean 8,000, SD 2,000, minimum 1,000              | Workforce implementation cost                    |
| Manual triage cost avoided            | GBP 20,000  | Triangular(0; 20,000; 40,000)                           | Operational cost offset                          |
| Extra review per false-priority study | 1.5 minutes | Normal mean 1.5, SD 0.5, minimum 0.1                    | Priority-review burden                           |

| Parameter                         | Base value          | Sensitivity or PSA specification | Role in the model                               |
|-----------------------------------|---------------------|----------------------------------|-------------------------------------------------|
| Review cost per minute            | GBP 1.20            | Fixed                            | Converts review burden to cost                  |
| Base willingness-to-pay threshold | GBP 30,000 per QALY | Scenario at GBP 50,000           | Converts health effects to net monetary benefit |

### Queue and Capacity Model

The queue model assumes that studies arrive throughout each working day and are read from a pooled worklist. The comparator uses first-in, first-out ordering. The intervention places flagged studies ahead of routine studies while preserving first-in, first-out ordering within each class. Service is non-pre-emptive: a study already being interpreted is not interrupted when a new priority study arrives.

The expected priority proportion is

where  $P()$  is prevalence,  $(Se)$  is sensitivity, and  $(Sp)$  is specificity. In the base case,

$$[ P() = (0.008)(0.90) + (0.992)(0.08) = 0.0866, ]$$

so approximately 8.7% of examinations enter the priority queue.

The synthetic DES used 200 replications of 1,000 working days. The first 500 days of each replication were discarded as warm-up. Random arrivals, variable daily capacity, and temporary capacity loss were represented explicitly. These settings are intended to demonstrate good simulation practice rather than reproduce a particular service.

| Assumption              | Worked specification                                                                      |
|-------------------------|-------------------------------------------------------------------------------------------|
| Arrival process         | Poisson arrivals with a mean of 100 examinations per working day                          |
| Reading capacity        | Normally distributed daily capacity with mean 101.4 and standard deviation 7 examinations |
| Capacity disruption     | A 5% daily probability of a 25% reduction in reading capacity                             |
| Comparator discipline   | First-in, first-out                                                                       |
| Intervention discipline | Priority studies first; first-in, first-out within priority and routine classes           |
| Pre-emption             | None                                                                                      |
| Additional capacity     | None; the intervention changes ordering rather than staffing or equipment                 |
| Warm-up                 | First 500 days discarded                                                                  |
| Replications            | 200 independent replications                                                              |

| Assumption             | Worked specification                                                        |
|------------------------|-----------------------------------------------------------------------------|
| Primary queue outcomes | Mean and 90th-percentile time to report by cancer status and priority class |

The queue results in the following table show the central trade-off. Prioritization substantially reduces reporting time for cancer cases but slightly increases waiting among routine studies. The overall mean changes little because the intervention redistributes capacity rather than adding capacity.

Table: Illustrative queue-model outputs

| Output                                 | Comparator     | AI priority | Interpretation                                               |
|----------------------------------------|----------------|-------------|--------------------------------------------------------------|
| Mean time to report, all studies       | 5.0 days       | 5.0 days    | Reordering does not create additional capacity               |
| 90th percentile, all studies           | 6.7 days       | 7.2 days    | The upper tail for routine studies becomes slightly longer   |
| Mean time to report, cancer cases      | 5.0 days       | 1.0 day     | Correct prioritization produces the main potential benefit   |
| Mean time, true-positive cancer cases  | 5.0 days       | 0.5 day     | Correctly flagged cases are read first                       |
| Mean time, false-negative cancer cases | 5.0 days       | 5.44 days   | Missed cases remain in the routine queue and may wait longer |
| Mean time, routine studies             | 5.0 days       | 5.44 days   | Priority cases impose an opportunity cost on routine work    |
| Priority share                         | Not applicable | 8.7%        | Includes true positives and false positives                  |

### Priority Classification and Consequences

For an annual cohort of size (N), the expected classification counts are

The base-case values produce 180 true positives, 20 false negatives, 1,984 false positives, and 22,816 true negatives. These categories matter because a priority system can create both benefit and opportunity cost even when it does not change the final diagnostic interpretation.

| Category      | Expected count | Queue consequence                  | Clinical and economic consequence                                                 |
|---------------|----------------|------------------------------------|-----------------------------------------------------------------------------------|
| True positive | 180            | Read rapidly in the priority queue | Potential reduction in materially delayed diagnosis and downstream treatment cost |

| Category       | Expected count | Queue consequence              | Clinical and economic consequence                                                      |
|----------------|----------------|--------------------------------|----------------------------------------------------------------------------------------|
| False negative | 20             | Remains in the routine queue   | Principal safety concern; delayed interpretation may reduce health benefit             |
| False positive | 1,984          | Uses priority-reading capacity | Additional review burden and longer routine waiting, without direct diagnostic benefit |
| True negative  | 22,816         | Remains in the routine queue   | Small opportunity cost from reordering; no direct diagnostic benefit                   |

### Time to Diagnosis and Long-Term Consequences

A fully developed evaluation would estimate disease-specific relationships among reporting time, diagnostic confirmation, treatment initiation, stage, survival, quality of life, and cost. To keep the worked example transparent, a synthetic relationship is used instead. The probability that a cancer case experiences a materially consequential delay is

where (d) is mean time to report in working days. The function is pedagogical; it is not a clinically validated delay model.

| Time to report | Material-delay probability | Interpretation                                        |
|----------------|----------------------------|-------------------------------------------------------|
| 1 day or less  | 0.04                       | Residual delay remains possible after rapid reporting |
| 3 days         | 0.08                       | Moderate increase in expected delay consequence       |
| 5 days         | 0.12                       | Base-case comparator region                           |
| 7 days         | 0.16                       | Greater likelihood of lost clinical benefit           |
| 9 days or more | 0.20                       | Probability capped in the illustrative function       |

Expected materially delayed cancer cases are

for the comparator and

for the intervention. Substitution of the base-case values gives

$$[ D_0 = (25,000)(0.008)(0.1202) = 24.04, ]$$

and

$$[ D_1 = (25,000)(0.008) = 9.78. ]$$

The model therefore estimates 14.26 materially delayed diagnoses avoided in one annual cohort.

### Costs and Outcomes

If (q) is the QALY loss per material delay and (c\_d) is the associated downstream cost, the incremental health effect is

With (q=0.10), the base-case gain is

$$[ E = (24.04 - 9.78)(0.10) = 1.43 . ]$$

False-priority review costs are calculated as

$$[ C_{\{FP\}} = (1,984)(1.5)(1.20) = 3,571. ]$$

The incremental cost is

The base-case calculation is

$$[ C = 65,000 + 20,000 + 18,000 + 8,000 + 3,571 - 20,000 - 35,657 = 58,914. ]$$

The incremental cost-effectiveness ratio is

Net monetary benefit at threshold ( $\lambda$ ) is

At GBP 30,000 per QALY, NMB is approximately GBP -16,125; at GBP 50,000 per QALY, it is approximately GBP 12,401. The base case is therefore not attractive at the lower threshold but becomes favourable at the higher threshold.

| Output                                     | Result      | Interpretation                                                        |
|--------------------------------------------|-------------|-----------------------------------------------------------------------|
| Materially delayed diagnoses, comparator   | 24.04       | Expected count in one annual cohort                                   |
| Materially delayed diagnoses, intervention | 9.78        | Weighted across true-positive and false-negative pathways             |
| Material delays avoided                    | 14.26       | Principal clinical benefit mechanism                                  |
| Incremental QALYs                          | 1.43        | Lifetime consequence assigned to the annual cohort                    |
| Gross programme cost                       | GBP 114,571 | Licence, integration, monitoring, training, and false-priority review |
| Manual triage cost avoided                 | GBP 20,000  | Operational offset                                                    |
| Downstream cost avoided                    | GBP 35,657  | Consequence of fewer material delays                                  |
| Net incremental cost                       | GBP 58,914  | Gross cost less offsets                                               |

| Output                     | Result              | Interpretation                                                |
|----------------------------|---------------------|---------------------------------------------------------------|
| ICER                       | GBP 41,305 per QALY | Above a GBP 30,000 threshold;<br>below a GBP 50,000 threshold |
| NMB at GBP 30,000 per QALY | GBP -16,125         | Unfavourable base-case decision                               |
| NMB at GBP 50,000 per QALY | GBP 12,401          | Favourable but uncertain base-case decision                   |

### Subgroup Assumptions and Equity

Average performance can conceal unequal access, different case complexity, and unequal waiting. Two synthetic subgroups are used to demonstrate this problem. They are not proxies for any protected characteristic and should be replaced by locally relevant clinical, demographic, geographic, disability, language, or access groups.

Table: Synthetic subgroup assumptions and outputs

| Measure                            | Group A  | Group B   | Difference  | Why it matters                                                          | Decision response                                |
|------------------------------------|----------|-----------|-------------|-------------------------------------------------------------------------|--------------------------------------------------|
| Share of examinations              | 80%      | 20%       | –           | Different service volume                                                | Preserve subgroup reporting                      |
| Cancer prevalence                  | 0.7%     | 1.2%      | +0.5 points | Higher expected need in Group B                                         | Use subgroup-specific prevalence                 |
| Priority sensitivity               | 92%      | 82%       | -10 points  | More false negatives in Group B                                         | Investigate data, threshold, and workflow causes |
| Priority specificity               | 93%      | 88%       | -5 points   | More false priorities in Group B                                        | Monitor review burden and calibration            |
| Baseline cancer report time        | 4.2 days | 6.2 days  | +2.0 days   | Existing access difference                                              | Do not interpret equal average benefit as equity |
| AI cancer report time              | 0.84 day | 1.88 days | +1.04 days  | Residual disparity remains after prioritization                         | Set subgroup time-to-report conditions           |
| False priorities per 1,000 studies | 69.5     | 118.6     | +49.1       | Greater capacity burden in Group B                                      | Test alternative thresholds and capacity         |
| QALYs gained per 1,000 studies     | 0.041    | 0.100     | +0.059      | Higher potential benefit reflects higher need and longer baseline delay | Report absolute and relative effects together    |

The example shows that a group can gain more in absolute terms yet still experience poorer performance and longer reporting times. Equity assessment should therefore consider baseline disadvantage, model performance, access to follow-up, absolute benefit, false-priority burden, and residual disparity rather than relying on one average metric.

### Deterministic Sensitivity Analysis

One-way sensitivity analysis changes one assumption at a time while holding the others at their base values. The base-case NMB at GBP 30,000 per QALY is GBP -16,125. The following table identifies the parameters most capable of changing that conclusion.

Table: Deterministic sensitivity analysis: net monetary benefit at GBP 30,000 per QALY

| Parameter                    | Low value  | High value | NMB at low  | NMB at high |
|------------------------------|------------|------------|-------------|-------------|
| Annual licence               | GBP 40,000 | GBP 90,000 | GBP 8,875   | GBP -41,125 |
| Priority sensitivity         | 0.80       | 0.95       | GBP -25,900 | GBP -11,238 |
| Priority specificity         | 0.88       | 0.96       | GBP -17,911 | GBP -14,339 |
| Baseline report time         | 3.0 days   | 7.0 days   | GBP -60,346 | GBP 27,654  |
| QALY loss per material delay | 0.05       | 0.15       | GBP -37,519 | GBP 5,270   |
| Cost per material delay      | GBP 1,000  | GBP 5,000  | GBP -37,519 | GBP 19,532  |
| Manual triage savings        | GBP 0      | GBP 40,000 | GBP -36,125 | GBP 3,875   |

The result is most sensitive to baseline delay, the clinical and economic consequences assigned to delay, programme price, and whether existing manual triage costs are genuinely avoided. Sensitivity and specificity matter, but their influence is smaller within the ranges examined. This pattern directs attention to local workflow and outcome evidence rather than to model accuracy alone.

### Probabilistic Sensitivity Analysis

The PSA jointly samples uncertain parameters and recalculates costs, QALYs, and NMB. The illustrative analysis used 200,000 Monte Carlo samples. Correlation among parameters was not represented, which is a limitation; local analysis should model known dependence among prevalence, performance, delay, costs, and outcomes. Transparent reporting of AI-specific economic evaluation and value-of-information analysis should follow appropriate reporting guidance (3).

| Output                     | Mean        | 95% simulation interval    |
|----------------------------|-------------|----------------------------|
| Incremental cost           | GBP 59,096  | GBP 14,894 to GBP 97,789   |
| Incremental QALYs          | 1.421       | 0.416 to 2.832             |
| NMB at GBP 30,000 per QALY | GBP -16,461 | GBP -74,216 to GBP 54,628  |
| NMB at GBP 50,000 per QALY | GBP 11,962  | GBP -62,610 to GBP 107,281 |

| Output                                            | Mean  | 95% simulation interval |
|---------------------------------------------------|-------|-------------------------|
| Probability cost-effective at GBP 30,000 per QALY | 28.8% | –                       |
| Probability cost-effective at GBP 50,000 per QALY | 57.8% | –                       |

The PSA does not support a confident unrestricted-adoption decision. At the lower threshold, most simulations favour the comparator. At the higher threshold, the intervention is more likely than not to be cost-effective, but uncertainty remains substantial.

## Value of Information

For strategies (j), the expected value of perfect information is

EVPI is the maximum expected benefit of eliminating all current parameter uncertainty before deciding. It is not the amount that should automatically be spent on research; research cost, delay, feasibility, and the number of future patients affected must also be considered. Reporting should follow transparent value-of-information guidance (4).

| Threshold           | Population EVPI | EVPI per examination | Interpretation                                                                 |
|---------------------|-----------------|----------------------|--------------------------------------------------------------------------------|
| GBP 30,000 per QALY | GBP 6,817       | GBP 0.27             | Limited value for an expensive definitive study serving only one annual cohort |
| GBP 50,000 per QALY | GBP 11,468      | GBP 0.46             | Greater value because the decision is closer to equipoise                      |

The most useful additional evidence would concern the local relationship between reporting delay and patient outcomes, priority performance in clinically and socially relevant subgroups, queue behaviour during capacity disruption, and the extent to which manual triage work is truly removed. Licence price is influential, but it is primarily a negotiation variable rather than a research parameter. A low-cost silent or controlled pilot may therefore be more proportionate than a large new accuracy study.

## Interpretation

The hybrid model reaches four main conclusions. - Prioritization can reduce reporting delay for cancer cases without changing total reading capacity, but the gain is partly financed by longer waiting among routine studies. - False negatives are the principal clinical risk, whereas false positives create review burden and an opportunity cost for the routine queue. - The base case is not cost-effective at GBP 30,000 per QALY and is only moderately favourable at GBP 50,000 per QALY. Price, baseline delay, delay-related consequences, and realised operational savings are more decision-critical than small changes in specificity. - Average results are insufficient. The subgroup example shows that higher absolute benefit may coexist with poorer sensitivity, more false priorities, and longer residual delay.

The model is intentionally simplified. It omits disease-stage transitions, treatment-capacity constraints, diagnostic anxiety, reader learning, model drift, material updates, and correlations among uncertain parameters. These omissions should be addressed when they could change the institutional decision. Credible modelling requires explicit context of use, version control, verification, validation, limitations, and independent review proportionate to consequence (5).

## Recommendation and Decision Conditions

The worked result does not support immediate unrestricted adoption at the base-case price and a GBP 30,000-per-QALY threshold. A controlled local pilot is more defensible because the intervention may create value if local delays are substantial, subgroup performance is acceptable, manual triage costs are removed, and price exposure is reduced.

At the base-case assumptions, the maximum annual licence compatible with zero NMB at GBP 30,000 per QALY is approximately GBP 48,875. At GBP 50,000 per QALY, the corresponding threshold is approximately GBP 77,401. These values are negotiation thresholds derived from the worked assumptions, not market prices.

| Domain                  | Condition                                                                                                                      | Decision response if not met                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Clinical performance    | Local sensitivity and calibration are acceptable for the intended pathway and relevant subgroups                               | Recalibrate, change threshold, restrict use, or stop                |
| Operational performance | Mean cancer reporting time is no more than 1.5 days, while routine mean and 90th-percentile delay remain within agreed limits  | Add capacity, redesign the queue, restrict priority volume, or stop |
| Safety                  | Every false-negative cancer case and any materially delayed diagnosis is reviewed                                              | Investigate, contain, revise threshold, and reassess intended use   |
| Equity                  | Subgroup sensitivity, false-priority burden, access to follow-up, and time to report meet predefined conditions                | Mitigate, redesign, restrict, or do not scale                       |
| Economic value          | Price and realised offsets produce non-negative NMB at the institution's approved threshold                                    | Renegotiate, reduce scope, or do not adopt                          |
| Governance              | Human interpretation, audit logs, version control, incident response, rollback, and named accountability remain operational    | Pause use until controls are restored                               |
| Evidence review         | Results are reviewed after six and twelve months using a locked protocol and archived model version                            | Continue only with documented approval and conditions               |
| Exit                    | Data export, workflow restoration, contract termination, and communication arrangements are tested before dependency increases | Do not scale until exit capability is credible                      |

The final decision should be recorded as one of the following: do not proceed; redesign and repeat; conduct a controlled pilot; adopt with conditions; scale; restrict; pause; replace; or retire. The recommendation should identify the model version, evidence cut-off, accountable owner, unresolved uncertainty, next review date, and stop criteria.

### Recommended Companion Files

For a reproducible digital release, this printable narrative should be distributed with: - a parameter workbook in XLSX format; - synthetic input data in CSV and JSON formats; - a standard Python script and a Jupyter notebook

implementing the queue, economic, subgroup, PSA, and EVPI calculations; - machine-readable model outputs and a static decision summary; - a data dictionary, model specification, test log, software-environment file, random seed, and version history; and - a blank institutional adaptation template that separates local evidence from the synthetic worked values.

## Supplemental Resources

These supplemental resources translate the Chapter 8 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 8 supplemental-resource set

| Resource                                      | Purpose and minimum content                                                                              | Expected output                                       |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Model Specification                           | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Verification, Validation, and Reproducibility | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Uncertainty and Value-of-Information Record   | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Model Specification*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Verification, Validation, and Reproducibility*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Specify the comparator, human reading workflow, priority threshold, queue discipline, and downstream pathway. - Use a classification decision tree to estimate true-positive, false-negative, false-positive, and true-negative consequences. - Use discrete-event simulation or a queueing model to estimate time-to-report distributions under constrained reader capacity. - Link material diagnostic delay to downstream cost and QALY consequences. - Represent subgroup prevalence, performance, access, and delay distributions explicitly. - Conduct deterministic sensitivity analysis, probabilistic sensitivity analysis, and value-of-information analysis.

### *Uncertainty and Value-of-Information Record*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### *Key Terms*

Table: Key terms for Chapter 8

| Term                 | Working definition and decision relevance                                                  |
|----------------------|--------------------------------------------------------------------------------------------|
| Decision model       | An explicit simplification of a decision problem and its causal and operational structure. |
| Simulation           | Computational representation of events, histories, queues, actors, or feedback.            |
| Value of information | The expected benefit of reducing uncertainty before or during commitment.                  |

### Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 8

| Instrument             | Status or type         | Appropriate use and limitation                       |
|------------------------|------------------------|------------------------------------------------------|
| Decision tree          | Model structure        | Suitable for short, nonrecurrent pathways.           |
| State-transition model | Model structure        | Represents repeated movement among health states.    |
| Microsimulation        | Individual-level model | Represents heterogeneity and history.                |
| DES                    | Simulation method      | Represents queues, resources, events, and timing.    |
| ABM/system dynamics    | Simulation methods     | Represent interactions or aggregate feedback.        |
| CHEERS-VOI             | Reporting guideline    | Supports reporting of value-of-information analysis. |

### Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                          | Purpose                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-B05 — Model Selection, Uncertainty, and Reproducibility<br>Core; docx; xlsx; csv; json; md; mmd; svg; png    | Select a decision model from the problem structure and document evidence, verification, uncertainty, and reproducibility.                       |
| AIHE-CASE08 — Imaging Prioritization: Decision Model and Value of Information<br>Worked case; md; csv; json; xlsx | Which uncertainty is most likely to change the adoption decision, and is further evidence worth its cost?                                       |
| AIHE-CASE08B — EVSI and Expected Net Benefit of Sampling for a Staged Pilot<br>Worked case; md; csv; json; xlsx   | Is an additional staged study worthwhile after accounting for expected information value, research cost, implementation delay, and sample size? |

| Resource                                                                                                                                            | Purpose                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-RG05 — Scenario Planning and Real-Options Worksheet<br>Reference guide; docx; xlsx; csv; json; md                                              | Examine uncertainty through alternative futures, thresholds, staged commitments, learning options, contract flexibility, and explicit exit criteria.                                                      |
| AIHE-AF — Evidence, Reporting, and Assurance Crosswalk<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                    | Match evidence and assurance needs to reporting standards, appraisal tools, lifecycle guidance, and management systems.                                                                                   |
| AIHE-EX04 — Evidence and Assumption Log<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                                   | Link each material claim and assumption to evidence, quality, applicability, uncertainty, owner, and action.                                                                                              |
| AIHE-EX06 — Model Card and AI Intervention Fact Sheet<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                     | Record the exact system, version, intended and prohibited uses, evidence, limitations, human oversight, and monitoring.                                                                                   |
| AIHE-EX15 — Analytic and Simulation Method Selection Guide<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                | Select decision trees, Markov models, microsimulation, discrete-event simulation, system dynamics, or agent-based modeling by problem structure.                                                          |
| AIHE-EX16 — Reproducible Analysis, Model Risk, and Code Quality Checklist<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Verify data provenance, environments, versioning, testing, validation, uncertainty, security, release, and archive controls.                                                                              |
| AIHE-PY14 — Deterministic sensitivity and tornado analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                 | Tests how one uncertain input at a time changes net monetary benefit or another linear decision outcome and ranks decision drivers.                                                                       |
| AIHE-PY15 — Probabilistic sensitivity analysis, CEAC, and EVPI<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                             | Propagates uncertainty in incremental costs and effects through Monte Carlo simulation, estimates cost-effectiveness probabilities across thresholds, and computes expected value of perfect information. |
| AIHE-PY18 — Decision-tree pathway model<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                                    | Calculates expected cost, effect, events, and net monetary benefit for short AI and comparator pathways with mutually exclusive outcome branches.                                                         |
| AIHE-PY19 — Markov cohort model for long-term AI evaluation<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                | Simulates repeated transitions among health states for AI and usual care, accumulating discounted costs and QALYs over time.                                                                              |
| AIHE-PY20 — Patient-level microsimulation<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                                  | Simulates heterogeneous patients over repeated cycles so history, subgroup risk, adherence, and individual outcomes can affect AI value.                                                                  |

## Chapter 9: Problem-First Procurement and Vendor Governance

### Chapter Orientation

Chapter 9 translates the problem, evidence, economics, safeguards, and implementation requirements into procurement and continuing vendor governance. The resources support pre-procurement specification, due diligence, non-compensatory safeguards, weighted comparison, update control, audit rights, pricing, continuity, renegotiation, and exit.

Use this suite before tendering, during vendor evaluation, at contract negotiation, and when renewal or renegotiation is required. The records are designed to reduce information asymmetry and to preserve local evidence access, monitoring, portability, rollback, and decommissioning options.

### Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                                  | How to use it                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Examples: Vendor Procurement; Upstream Model-Provider Change                                               | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Pre-Procurement Requirements; Vendor Scorecard and Mandatory Safeguards; Contract, Renegotiation, and Exit Record | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | Problem-first procurement, Mandatory safeguard, Vendor governance                                                 | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |
| Repository coverage    | 1 core, 2 Python analytics, 3 supplemental specialist tools, 2 worked cases                                       | Use the catalog map below to locate editable, process-flow, data, and computational variants.                  |

### Worked Example: Vendor Procurement and Renegotiation

#### *Purpose and Status*

A hospital is procuring a clinically consequential AI platform. Three vendors differ in evidence quality, interoperability, pricing, monitoring access, update control, and exit support. A prior procurement also requires renegotiation because implementation costs and vendor restrictions exceeded the original assumptions.

#### *Decision Problem and Comparators*

Decision problem. Which vendor, if any, should proceed to a controlled contract, and what remedies or exit actions are required for the existing platform?

Table: Comparators for the Chapter 9 worked example

| Option | Comparator or strategy                           |
|--------|--------------------------------------------------|
| 1      | Vendor A: strong evidence, high fixed price      |
| 2      | Vendor B: lower price, limited monitoring access |

| Option | Comparator or strategy                                          |
|--------|-----------------------------------------------------------------|
| 3      | Vendor C: shared-service model with stronger exit terms         |
| 4      | Internal or non-AI process alternative                          |
| 5      | Renegotiate, replace, restrict, or retire the existing platform |

### *Synthetic Parameters and Evidence Sources*

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 9 worked example

| Parameter                                    | Base value       | Source or interpretation  |
|----------------------------------------------|------------------|---------------------------|
| Mandatory safeguard domains                  | 7                | Procurement specification |
| Weighted domains                             | 7                | Local scorecard           |
| Expected annual volume                       | 180,000 uses     | Activity forecast         |
| Maximum acceptable variable exposure         | EUR 420,000/year | Budget condition          |
| Required update notice                       | 90 days          | Contract requirement      |
| Minimum data-export period after termination | 12 months        | Continuity requirement    |
| Remediation period for existing vendor       | 120 days         | Renegotiation proposal    |

### *Step-by-Step Analysis*

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                  | Decision record                               |
|------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Define the service problem, comparator set, intended and excluded uses, and measurable outcomes.               | Evidence and responsible owner to be recorded |
| 2    | Require version-matched evidence and complete technical, data, security, economic, and workflow due diligence. | Evidence and responsible owner to be recorded |

| Step | Required work                                                                                                                                                                   | Decision record                               |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 3    | Apply mandatory safeguards before weighted comparison.                                                                                                                          | Evidence and responsible owner to be recorded |
| 4    | Model pricing and lifecycle cost under realistic volume, update, support, and exit scenarios.                                                                                   | Evidence and responsible owner to be recorded |
| 5    | Convert monitoring, audit, update, continuity, portability, and exit requirements into enforceable clauses.                                                                     | Evidence and responsible owner to be recorded |
| 6    | For the existing platform, compare prospective continuation, renegotiation, restriction, replacement, and decommissioning without allowing sunk cost to determine the decision. | Evidence and responsible owner to be recorded |

### *Illustrative Outputs*

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                      | Result                                                | Interpretation                                                     |
|-----------------------------|-------------------------------------------------------|--------------------------------------------------------------------|
| Vendor B weighted score     | Highest before safeguards                             | Ineligible because monitoring access fails a mandatory requirement |
| Vendor A evidence profile   | Strongest                                             | Price and exit terms require negotiation                           |
| Vendor C continuity profile | Strongest                                             | Local evidence and governance arrangements require confirmation    |
| Existing vendor decision    | Time-limited remediation with parallel exit plan      | Continuation is conditional                                        |
| Preferred award approach    | Competitive dialogue followed by conditional contract | No automatic award from aggregate score                            |

### *Interpretation and Recommendation*

Proceed only with a vendor that passes every mandatory safeguard and accepts version-specific evidence, monitoring access, transparent pricing, material-change control, continuity, and exit obligations. Continue the existing platform only under a time-limited remediation agreement and maintain a funded replacement plan.

- mandatory safeguards are evaluated independently of weighted preferences.
- missing evidence receives clarification, conservative treatment, or exclusion.
- contracted pricing includes integration, monitoring, updates, incident support, and exit.
- the institution retains audit, data, log, and publication rights needed for accountability.

- renewal is linked to realized value and current evidence rather than sunk cost.

## Supplemental Resources

These supplemental resources translate the Chapter 9 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 9 supplemental-resource set

| Resource                                  | Purpose and minimum content                                                                              | Expected output                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Pre-Procurement Requirements              | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Vendor Scorecard and Mandatory Safeguards | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Contract, Renegotiation, and Exit Record  | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Pre-Procurement Requirements*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Vendor Scorecard and Mandatory Safeguards*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define the service problem, comparator set, intended and excluded uses, and measurable outcomes. - Require version-matched evidence and complete technical, data, security, economic, and workflow due diligence. - Apply mandatory safeguards before weighted comparison. - Model pricing and lifecycle cost under realistic volume, update, support, and exit scenarios. - Convert monitoring, audit, update, continuity, portability, and exit requirements into enforceable clauses. - For the existing platform, compare prospective continuation, renegotiation, restriction, replacement, and decommissioning without allowing sunk cost to determine the decision.

### *Contract, Renegotiation, and Exit Record*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### *Key Terms*

Table: Key terms for Chapter 9

| Term                      | Working definition and decision relevance                                                   |
|---------------------------|---------------------------------------------------------------------------------------------|
| Problem-first procurement | A process that defines need, comparator, evidence, and safeguards before product selection. |
| Mandatory safeguard       | A non-compensatory requirement that must be met before weighted comparison.                 |
| Vendor governance         | Continuing oversight of evidence, performance, change, continuity, and exit after award.    |

### Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 9

| Instrument                        | Status or type                            | Appropriate use and limitation                                       |
|-----------------------------------|-------------------------------------------|----------------------------------------------------------------------|
| NICE Evidence Standards Framework | Evidence guidance                         | Supports evidence planning for digital health technologies.          |
| DTAC                              | Jurisdiction-specific assessment criteria | Supports baseline digital-technology due diligence in England.       |
| EU model contractual clauses      | Public procurement resource               | Provides adaptable AI contractual clauses.                           |
| FAIR-AI                           | Implementation and review framework       | Links pre-implementation review with post-implementation monitoring. |

### Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                    | Purpose                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-D01 — Vendor Evidence and Contract Requirements<br>Core; docx; xlsx; csv; json; md; mmd; svg; png      | Translate evidence, interoperability, data, security, monitoring, update, service, remedy, portability, and exit requirements into contract terms. |
| AIHE-CASE09 — AI Vendor Procurement and Contract Negotiation<br>Worked case; md; csv; json; xlsx            | How should the hospital compare vendors when a lower price is accompanied by weak audit, security, update, and exit provisions?                    |
| AIHE-CASE09B — Upstream Model-Provider Change During a Vendor Contract<br>Worked case; md; csv; json; xlsx  | Does a change in the underlying model or hosting dependency require disclosure, revalidation, contractual action, and reauthorization?             |
| AIHE-D — AI Procurement Checklist<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Evaluate problem fit, evidence, integration, lifecycle cost, data rights, security, monitoring, updates, audit, and exit.                          |

| Resource                                                                                                                      | Purpose                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-EX08 — Vendor Demonstration Evaluation Form<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png    | Structure demonstrations around real workflows, evidence, limitations, usability, interoperability, monitoring, and unsupported claims.                          |
| AIHE-EX09 — Contract Requirements and Exit Matrix<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png   | Document intended use, evidence, audit, updates, data, security, service levels, pricing, termination, portability, and decommissioning clauses.                 |
| AIHE-PY17 — AI pricing and contract scn analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                     | Compares subscription, per-use, risk-sharing, and hybrid pricing arrangements under different volume, uptake, performance, and savings assumptions.              |
| AIHE-PY25 — Weighted procurement scorecard with mandatory safeguards<br>Python analytics; py; ipynb; xlsx; csv; json; png; md | Ranks candidate AI products using weighted evidence domains while enforcing non-compensatory safety, privacy, equity, cybersecurity, audit, and exit safeguards. |

## Part IV: Governance, Evidence, and Lifecycle Decision-Making

This Part brings governance, accountability, evidence, validation, lifecycle operations, and continuing decision-making into one institutional system.

### Chapter 10: Governance, Accountability, and Institutional Risk

#### Chapter Orientation

Chapter 10 integrates governance, accountability, privacy, cybersecurity, equity, digital inclusion, trust, and institutional risk. The resources make decision rights explicit and prevent favourable expected value from compensating for an unacceptable safeguard failure.

Use this suite for patient-facing systems, consequential decision support, public communication, and any intervention whose risks extend beyond technical performance. The outputs should identify affected groups, lawful authority, meaningful human routes, accessibility, incident response, residual risk, and the authority to restrict or stop use.

#### Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                                 | How to use it                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Patient-Facing Chatbot and Digital Inclusion                                                     | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Governance and Accountability Record; Integrated Institutional-Risk Register; Patient-Facing AI Assurance Record | Create the institutional records needed for analysis, approval, monitoring, and review.                        |

| Layer               | Chapter resource                                                                             | How to use it                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Reference focus     | Governance, Accountability, Institutional risk                                               | Clarify terminology, method selection, instrument status, formulas, and update requirements.  |
| Repository coverage | 2 core, 3 python analytics, 1 reference guide, 6 supplemental specialist tool, 1 worked case | Use the catalog map below to locate editable, process-flow, data, and computational variants. |

## Worked Example: Patient-Facing Chatbot and Digital Inclusion

### Purpose and Status

A hospital proposes a multilingual patient-facing chatbot for appointment guidance, preparation instructions, and symptom navigation. The intervention could improve access but may also create factual, privacy, cybersecurity, accessibility, workload, and trust risks.

### Decision Problem and Comparators

Decision problem. Should the chatbot be piloted, restricted to administrative tasks, redesigned with human escalation, or rejected in favour of non-AI navigation improvements?

Table: Comparators for the Chapter 10 worked example

| Option | Comparator or strategy                         |
|--------|------------------------------------------------|
| 1      | Improved static web and printed information    |
| 2      | Call-centre redesign and multilingual staffing |
| 3      | Rules-based conversational interface           |
| 4      | Restricted AI chatbot with human escalation    |

### Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 10 worked example

| Parameter                                        | Base value      | Source or interpretation |
|--------------------------------------------------|-----------------|--------------------------|
| Languages proposed                               | 8               | Service plan             |
| Share of users requiring assisted digital access | 18%             | Patient survey           |
| Queries classified as clinically consequential   | 24%             | Retrospective sample     |
| Human escalation target                          | Under 5 minutes | Safety requirement       |

| Parameter                     | Base value | Source or interpretation |
|-------------------------------|------------|--------------------------|
| Retention of interaction logs | 30 days    | Draft policy             |
| Expected annual interactions  | 420,000    | Activity forecast        |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                       | Decision record                               |
|------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Define allowed, prohibited, and escalation-required tasks.                                                          | Evidence and responsible owner to be recorded |
| 2    | Map data flows, third parties, retention, logging, and model-service dependencies.                                  | Evidence and responsible owner to be recorded |
| 3    | Assess factuality, information integrity, privacy, cybersecurity, language, accessibility, and digital inclusion.   | Evidence and responsible owner to be recorded |
| 4    | Estimate the effect on call-centre demand, repeat contacts, abandonment, and downstream services.                   | Evidence and responsible owner to be recorded |
| 5    | Assign accountability, incident communication, contestability, and stopping authority.                              | Evidence and responsible owner to be recorded |
| 6    | Pilot only where essential non-digital routes remain available and monitoring can detect differential use and harm. | Evidence and responsible owner to be recorded |

### Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                 | Result                                                                                  | Interpretation                    |
|------------------------|-----------------------------------------------------------------------------------------|-----------------------------------|
| Suitable initial scope | Administrative guidance and verified preparation instructions                           | Symptom advice remains restricted |
| Primary equity risk    | Exclusion of users with language, disability, literacy, connectivity, or trust barriers | Requires human alternatives       |

| Output                  | Result                                                  | Interpretation                                 |
|-------------------------|---------------------------------------------------------|------------------------------------------------|
| Primary governance risk | Unclear responsibility for harmful or incomplete advice | Requires explicit escalation and logs          |
| Expected call reduction | Uncertain                                               | Must account for repeat and escalated contacts |
| Preferred decision      | Restricted pilot with independent content verification  | Not unrestricted patient-facing deployment     |

### *Interpretation and Recommendation*

Pilot a restricted, clearly disclosed service for low-consequence tasks, preserve immediate human alternatives, and prohibit unsupported diagnosis or treatment advice. Expansion should depend on effective access and outcome evidence rather than interaction volume.

- all content domains have accountable clinical or operational owners.
- language and accessibility testing includes affected users.
- privacy, retention, and third-party data use are transparent and minimized.
- escalation reliability and abandonment are monitored by subgroup.
- a serious factual, privacy, security, or equity failure triggers immediate restriction and review.

### **Supplemental Resources**

These supplemental resources translate the Chapter 10 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 10 supplemental-resource set

| Resource                               | Purpose and minimum content                                                                              | Expected output                                       |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Governance and Accountability Record   | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Integrated Institutional-Risk Register | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Patient-Facing AI Assurance Record     | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Governance and Accountability Record*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Integrated Institutional-Risk Register*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define allowed, prohibited, and escalation-required tasks. - Map data flows, third parties, retention, logging, and model-service dependencies. - Assess factuality, information integrity, privacy, cybersecurity, language, accessibility, and digital inclusion. - Estimate the effect on call-centre demand, repeat contacts, abandonment, and downstream services. - Assign accountability, incident communication, contestability, and stopping authority. - Pilot only where essential non-digital routes remain available and monitoring can detect differential use and harm.

### *Patient-Facing AI Assurance Record*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## **Reference Resources**

### *Key Terms*

Table: Key terms for Chapter 10

| Term               | Working definition and decision relevance                                                         |
|--------------------|---------------------------------------------------------------------------------------------------|
| Governance         | Institutional decisions, authority, controls, records, and responses governing consequential use. |
| Accountability     | The obligation to justify decisions and accept responsibility for consequences.                   |
| Institutional risk | Risk to safety, rights, continuity, equity, trust, resources, and legitimacy.                     |

### *Status and Appropriate Use of Selected Instruments*

Table: Selected instruments relevant to Chapter 10

| Instrument                  | Status or type                                   | Appropriate use and limitation                                           |
|-----------------------------|--------------------------------------------------|--------------------------------------------------------------------------|
| EU AI Act                   | Binding EU regulation in scope                   | Creates obligations based on role and risk classification.               |
| EHDS                        | Binding EU regulation with phased implementation | Governs electronic health data use and infrastructure in scope.          |
| ISO/IEC 42001, 23894, 42005 | Voluntary international standards                | Support management systems, risk management, and impact assessment.      |
| NIST AI RMF                 | Voluntary risk-management framework              | Organizes Govern, Map, Measure, and Manage activities.                   |
| FUTURE-AI and FAIR-AI       | Research and consensus guidance                  | Support trustworthy implementation and review; not regulatory standards. |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                           | Purpose                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-C03 — Integrated Institutional-Risk Review<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                                  | Review ethics, equity, privacy, cybersecurity, safety, continuity, trust, legal, and economic risks as one institutional system.                                                |
| AIHE-C05 — Public Transparency and Communication Record<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                          | Define accurate, accessible communication about intended use, limitations, oversight, data, complaints, incidents, and alternatives.                                            |
| AIHE-CASE10 — Patient-Facing Chatbot Institutional-Risk Review<br>Worked case; md; csv; json; xlsx                                 | Which uses can be permitted when convenience and call reduction coexist with privacy, safety, equity, and communication risks?                                                  |
| AIHE-RG04 — Regulatory, Standards, and Policy Update Register<br>Reference guide; docx; xlsx; csv; json; md                        | Maintain date-stamped records for European, Romanian, and global legal, policy, standards, reimbursement, and guidance sources whose scope or implementation status may change. |
| AIHE-AB — Privacy and Cybersecurity Risk Register<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png        | Record privacy, security, continuity, supplier, model, and incident risks with controls, owners, and review dates.                                                              |
| AIHE-AC — Equity Impact Tool<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                             | Identify affected groups, barriers, harms, mitigations, costs, residual concerns, and monitoring indicators.                                                                    |
| AIHE-C — Ethical Governance Checklist for Healthcare AI<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png  | Translate autonomy, safety, transparency, fairness, oversight, accountability, and public value into evidence and action.                                                       |
| AIHE-EX13 — Public Transparency and Communication Notice<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Explain AI use, limits, human oversight, data handling, accessibility, questions, complaints, incidents, and human alternatives.                                                |
| AIHE-G — Sample Health-System AI Governance Charter<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png      | Establish a health-system governance charter covering purpose, scope, membership, evidence, decisions, monitoring, incidents, updates, and de-adoption.                         |
| AIHE-N — Trust, Equity, and Public Value Workbook<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png        | Integrate trust-adjusted value, distributional effects, mitigation, public value, confidence, and mandatory safeguards.                                                         |
| AIHE-PY07 — Feature importance and partial-dependence exploration<br>Python analytics; py; ipynb; xlsx; csv; json; png; md         | Fits a tree ensemble, calculates permutation importance, and visualizes the relationship between the top feature and predicted risk.                                            |

| Resource                                                                                                                                        | Purpose                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-PY08 — Subgroup performance and equity audit<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                      | Compares AUC, calibration, sensitivity, specificity, predictive values, and selection rates across population groups.                       |
| AIHE-PY26 — Integrated privacy, cybersecurity, equity, safety, and trust risk register<br>Python analytics; py; ipynb; xlsx; csv; json; png; md | Scores inherent and residual risk, checks mitigation completeness, prioritizes overdue actions, and produces an accountable risk dashboard. |

## Chapter 11: Evidence, Validation, and Real-World Evaluation

### Chapter Orientation

Chapter 11 addresses evidence maturity, appraisal, external and local validation, silent evaluation, causal evaluation, implementation research, mixed methods, and evidence synthesis. The resources distinguish model validity from the effect of using the complete intervention in practice.

Use this suite to design local validation, prospective evaluation, subgroup analysis, decision-curve analysis, causal studies, and decision-ready evidence records. The connected readmission case separates predictive performance from the effect of the care-management pathway and from the resources needed to deliver it.

### Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                                   | How to use it                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Readmission-Risk Prediction: Validation and Causal Evaluation                                      | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Evidence Maturity and Appraisal; Local Validation and Silent Evaluation; Causal and Real-World Evaluation Protocol | Create the institutional records needed for analysis, approval, monitoring, and review.                        |
| Reference focus        | Validation, Clinical utility, Causal evaluation                                                                    | Clarify terminology, method selection, instrument status, formulas, and update requirements.                   |
| Repository coverage    | 6 python analytics, 2 reference guide, 1 worked case                                                               | Use the catalog map below to locate editable, process-flow, data, and computational variants.                  |

### Worked Example: Readmission-Risk Prediction: Validation and Causal Evaluation

#### Purpose and Status

A hospital has acquired a readmission-risk model intended to direct care-management follow-up. External validation is favourable, but the local population, documentation, social-risk information, and care-management capacity differ from the development setting.

#### Decision Problem and Comparators

Decision problem. Is the exact model and intervention sufficiently valid and useful for local use, and does the targeted follow-up pathway cause better outcomes than current discharge planning?

Table: Comparators for the Chapter 11 worked example

| Option | Comparator or strategy                                        |
|--------|---------------------------------------------------------------|
| 1      | Current discharge planning                                    |
| 2      | Conventional risk checklist                                   |
| 3      | Model-guided follow-up with existing capacity                 |
| 4      | Model-guided follow-up with expanded care-management capacity |

### *Synthetic Parameters and Evidence Sources*

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 11 worked example

| Parameter                               | Base value         | Source or interpretation                       |
|-----------------------------------------|--------------------|------------------------------------------------|
| Annual eligible discharges              | 18,000             | Hospital activity                              |
| 30-day readmission prevalence           | 14%                | Baseline cohort                                |
| Available care-management slots         | 65/week            | Service capacity                               |
| Model sensitivity at proposed threshold | 0.76               | Local silent evaluation                        |
| Positive predictive value               | 0.31               | Local silent evaluation                        |
| Calibration slope                       | 0.82               | Local validation                               |
| Expected follow-up effect               | Relative risk 0.88 | Evidence synthesis; causal uncertainty remains |

### *Step-by-Step Analysis*

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                                     | Decision record                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Define the intended use, threshold, action, capacity, and exact system version.                                                   | Evidence and responsible owner to be recorded |
| 2    | Assess data design, leakage, transportability, discrimination, calibration, threshold-specific utility, and subgroup performance. | Evidence and responsible owner to be recorded |

| Step | Required work                                                                                                                            | Decision record                               |
|------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 3    | Conduct silent evaluation before clinical dependence.                                                                                    | Evidence and responsible owner to be recorded |
| 4    | Separate model validation from causal evaluation of the follow-up intervention.                                                          | Evidence and responsible owner to be recorded |
| 5    | Use an appropriate randomized, quasi-experimental, or target-trial design to estimate service effects.                                   | Evidence and responsible owner to be recorded |
| 6    | Combine technical, clinical, implementation, qualitative, economic, governance, and post-deployment evidence in a decision-ready record. | Evidence and responsible owner to be recorded |

### *Illustrative Outputs*

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output                     | Result                                             | Interpretation                             |
|----------------------------|----------------------------------------------------|--------------------------------------------|
| Local discrimination       | Acceptable                                         | Not sufficient for adoption                |
| Calibration                | Overprediction in higher-risk groups               | Requires recalibration or threshold review |
| Weekly patients flagged    | Approximately 120                                  | Exceeds current care-management capacity   |
| Causal effect of follow-up | Uncertain locally                                  | Requires controlled evaluation             |
| Preferred decision         | Recalibrate and conduct capacity-constrained pilot | No unrestricted implementation             |

### *Interpretation and Recommendation*

Complete recalibration and a prospective, capacity-aware evaluation of the model-plus-follow-up intervention. The model should not be judged effective merely because it predicts readmission; the evaluation must show that targeted action improves outcomes relative to realistic alternatives.

- silent-mode evidence and subgroup calibration are acceptable.
- the threshold fits available care-management capacity.
- the causal evaluation separates model performance from intervention effect.
- implementation costs and workload are measured prospectively.
- the evidence record states attribution limits and conditions for continuation.

## Supplemental Resources

These supplemental resources translate the Chapter 11 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 11 supplemental-resource set

| Resource                                  | Purpose and minimum content                                                                              | Expected output                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Evidence Maturity and Appraisal           | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Local Validation and Silent Evaluation    | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Causal and Real-World Evaluation Protocol | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

### *Evidence Maturity and Appraisal*

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

### *Local Validation and Silent Evaluation*

The following sequence converts findings into accountable action. The local record should name the responsible person for each activity, the completion date, the evidence of completion, and any residual uncertainty. - Define the intended use, threshold, action, capacity, and exact system version. - Assess data design, leakage, transportability, discrimination, calibration, threshold-specific utility, and subgroup performance. - Conduct silent evaluation before clinical dependence. - Separate model validation from causal evaluation of the follow-up intervention. - Use an appropriate randomized, quasi-experimental, or target-trial design to estimate service effects. - Combine technical, clinical, implementation, qualitative, economic, governance, and post-deployment evidence in a decision-ready record.

### *Causal and Real-World Evaluation Protocol*

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### *Key Terms*

Table: Key terms for Chapter 11

| Term       | Working definition and decision relevance                              |
|------------|------------------------------------------------------------------------|
| Validation | Assessment of performance and suitability for a stated context of use. |

| Term              | Working definition and decision relevance                                           |
|-------------------|-------------------------------------------------------------------------------------|
| Clinical utility  | Whether use of information improves decisions or outcomes relative to alternatives. |
| Causal evaluation | Estimation of what would have happened under a credible counterfactual.             |

### Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 11

| Instrument            | Status or type                               | Appropriate use and limitation                                              |
|-----------------------|----------------------------------------------|-----------------------------------------------------------------------------|
| TRIPOD+AI             | Reporting guideline                          | Prediction-model studies.                                                   |
| PROBAST+AI            | Appraisal tool                               | Risk of bias and applicability for prediction-model evidence.               |
| STARD-AI              | Reporting guideline                          | Diagnostic-accuracy studies.                                                |
| SPIRIT-AI/CONSORT-AI  | Reporting guidelines                         | Trial protocols and reports.                                                |
| DECIDE-AI             | Reporting guideline                          | Early-stage live clinical evaluation.                                       |
| MRC/CFIR/RE-AIM/PRISM | Methodological and implementation frameworks | Support programme theory, determinants, outcomes, context, and sustainment. |

### Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                     | Purpose                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-CASE11 — Local Validation and Causal Evaluation of a Readmission Model<br>Worked case; md; csv; json; xlsx              | Does local performance and observed service change support adoption, and what can reasonably be attributed to the AI-enabled pathway?                                                                                 |
| AIHE-RG02 — Reporting and Appraisal Tool Selection Guide<br>Reference guide; docx; xlsx; csv; json; md                       | Match evidence type and study stage to current reporting and appraisal instruments without treating checklist completion as proof of validity, effectiveness, or regulatory conformity.                               |
| AIHE-RG06 — Diagnostic and Predictive Evidence Appraisal Checklist<br>Reference guide; docx; xlsx; csv; json; md             | Appraise reference standards, index tests, data partitions, calibration, diagnostic accuracy, bias, subgroup performance, clinical utility, and downstream resource consequences.                                     |
| AIHE-EX05 — Local Validation Report Template<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png; html | Document intended use, local cohort, exact system version, discrimination, calibration, threshold-specific performance, subgroup findings, workflow, clinical utility, uncertainty, and the resulting local decision. |

| Resource                                                                                                                                       | Purpose                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-PY04 — Predictive model development: logistic regression and random forest<br>Python analytics; py; ipynb; xlsx; csv; json; png; md       | Builds two baseline classifiers, compares test performance, and exports predictions for later validation modules.                                                   |
| AIHE-PY05 — Model performance, threshold trade-offs, and calibration<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                  | Calculates ROC AUC, Brier score, confusion-matrix measures across thresholds, calibration intercept and slope, and calibration bins.                                |
| AIHE-PY24 — Implementation evaluation with interrupted time series and control charts<br>Python analytics; py; ipynb; xlsx; csv; json; png; md | Evaluates change after an AI implementation using segmented regression, run charts, control limits, uptake, and balancing measures.                                 |
| AIHE-PY30 — Random-effects meta-analysis and forest plot<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                              | Pools study-level effects using inverse-variance fixed and DerSimonian-Laird random-effects methods, quantifies heterogeneity, and creates forest and funnel plots. |
| AIHE-PY31 — Difference-in-differences evaluation<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                      | Estimates the effect of AI implementation by comparing change over time in an intervention group with change in a comparison group.                                 |
| AIHE-PY32 — Time-to-event and survival analysis<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                                       | Estimates Kaplan-Meier survival curves and compares event timing between AI and comparator groups using a log-rank test.                                            |

## Chapter 12: Decision Architecture and Lifecycle Governance

### Chapter Orientation

Chapter 12 brings specialist evidence into a defensible lifecycle decision architecture. The resources organize AI-HEVG assessment, the AI Health Economics Dossier, current institutional decision states, decision provenance, monitoring, incidents, material change, reauthorization, renewal, de-adoption, replacement, retirement, and the formal end of authorization.

Use this suite when several specialist analyses must be integrated without collapsing their findings into one score. The records preserve the exact intervention identity, current state, scope, conditions, mandatory safeguards, unresolved uncertainty, accountable authority, transition triggers, decision history, and next review.

### Resource Suite at a Glance

| Layer                  | Chapter resource                                                                                                  | How to use it                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Worked example         | Worked Example: Readmission-Risk Lifecycle Decision                                                               | Follow the complete decision pathway and identify which local evidence must replace the synthetic assumptions. |
| Supplemental resources | Integrated Assessment and AI-HED; Post-Deployment Monitoring and Incident Record; Accountable Adoption Principles | Create the institutional records needed for analysis, approval, monitoring, and review.                        |

| Layer               | Chapter resource                                                                                      | How to use it                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Reference focus     | Decision architecture, Algorithmovigilance, AI-HED                                                    | Clarify terminology, method selection, instrument status, formulas, and update requirements.  |
| Repository coverage | 4 core, 4 Python analytics, 1 reference guide, 15 active supplemental specialist tools, 1 worked case | Use the catalog map below to locate editable, process-flow, data, and computational variants. |

### Worked Example: Readmission-Risk Lifecycle Decision

#### Purpose and Status

Following local validation and a controlled pilot, the hospital must determine the intervention's current institutional decision state. The AI-HED contains model and intervention evidence, service effects, lifecycle cost, subgroup consequences, workload, supplier changes, conditions, safeguards, transition triggers, and unresolved uncertainty.

#### Decision Problem and Comparators

Decision problem. Which current decision state is justified: continued evidence development, restricted pilot, conditional use, routine authorized use, suspension, replacement, or retirement?

Table: Comparators for the Chapter 12 worked example

| Option | Comparator or strategy                       |
|--------|----------------------------------------------|
| 1      | Continue limited pilot                       |
| 2      | Scale with current threshold and capacity    |
| 3      | Scale with expanded care-management capacity |
| 4      | Restrict to selected services or subgroups   |
| 5      | Return to the conventional discharge pathway |

#### Synthetic Parameters and Evidence Sources

The following table provides the base-case inputs. A production analysis should add parameter distributions, correlations, structural alternatives, data dates, system versions, and evidence-confidence ratings.

Table: Synthetic parameters for the Chapter 12 worked example

| Parameter                      | Base value | Source or interpretation    |
|--------------------------------|------------|-----------------------------|
| Pilot duration                 | 9 months   | AI-HED record               |
| Readmission relative risk      | 0.91       | Local controlled evaluation |
| Incremental cost per discharge | EUR 94     | TCAIO record                |
| Care-management capacity used  | 93%        | Operations dashboard        |

| Parameter                         | Base value                                | Source or interpretation     |
|-----------------------------------|-------------------------------------------|------------------------------|
| Subgroup with calibration concern | Patients with incomplete social-risk data | distributional equity review |
| Material vendor update expected   | Within 6 months                           | Change notice                |

### Step-by-Step Analysis

The sequence below is intended to make the analysis reproducible and to ensure that evidence, economics, governance, and implementation develop together.

Table: Analytical sequence

| Step | Required work                                                                                                                          | Decision record                               |
|------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1    | Use AI-HEVG to integrate problem relevance, data readiness, clinical validity, economic value, ethics, and organizational feasibility. | Evidence and responsible owner to be recorded |
| 2    | Use AI-HED as the living decision record linking evidence, assumptions, decisions, and versions.                                       | Evidence and responsible owner to be recorded |
| 3    | Apply non-compensatory safeguards before considering aggregate scores.                                                                 | Evidence and responsible owner to be recorded |
| 4    | Assign accountable roles through the accountability matrix and define monitoring, incidents, material change, renewal, and exit.       | Evidence and responsible owner to be recorded |
| 5    | Use GATED-AI, VALIDATE-AI, and other processes only for their defined operational roles.                                               | Evidence and responsible owner to be recorded |
| 6    | Reach a decision with conditions, monitoring thresholds, review dates, and stopping authority.                                         | Evidence and responsible owner to be recorded |

### Illustrative Outputs

The outputs in the following table show how technical, clinical, operational, economic, equity, and governance findings should be kept visible rather than compressed into a single favourable or unfavourable score.

Table: Illustrative model and decision outputs

| Output          | Result               | Interpretation                                            |
|-----------------|----------------------|-----------------------------------------------------------|
| Clinical effect | Modest and uncertain | Direction favourable; magnitude below initial expectation |

| Output             | Result                                       | Interpretation                                         |
|--------------------|----------------------------------------------|--------------------------------------------------------|
| Economic value     | Conditional                                  | Dependent on capacity expansion and avoided admissions |
| Equity concern     | Material but potentially remediable          | Incomplete social-risk data affect prioritization      |
| Lifecycle concern  | Upcoming material update                     | Requires version-specific reassessment                 |
| Preferred decision | Continue with conditions; do not scale fully | Evidence and capacity not yet sufficient               |

### *Interpretation and Recommendation*

Maintain restricted, conditional use while care-management capacity and social-risk data are addressed and the proposed material update is evaluated. Movement to routine authorized use requires version-matched revalidation, revised economic and equity assessment, and a formally updated AI-HED.

- AI-HED is updated with the exact deployed version, evidence cut-off, and decision rationale.
- monitoring includes workload, outcomes, subgroup access, cost, incidents, and trust.
- the material update is assessed before deployment and rollback remains available.
- the accountable authority can restrict or retire use without vendor approval.
- renewal is based on current evidence and realized value rather than prior investment.

### **Supplemental Resources**

These supplemental resources translate the Chapter 12 concepts into editable institutional records. They should be adapted to the intended use and consequence, completed by a multidisciplinary team, and maintained under version control. They are not validated clinical, legal, regulatory, procurement, accreditation, or health technology assessment instruments.

Table: Chapter 12 supplemental-resource set

| Resource                                       | Purpose and minimum content                                                                              | Expected output                                       |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Integrated Assessment and AI-HED               | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Post-Deployment Monitoring and Incident Record | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |
| Accountable Adoption Principles                | Define scope, owner, evidence source, assumptions, decision threshold, review date, and required action. | Blank record, synthetic example, and version history. |

## Integrated Assessment and AI-HED

Complete the common institutional record described in the opening guide, then add the chapter-specific evidence and fields identified below. Explain any field judged not applicable; do not replace missing evidence with an implicit assumption.

## Post-Deployment Monitoring and Incident Record

The following sequence converts integrated evidence into a living institutional decision. - Use AI-HEVG to examine problem relevance, data readiness, clinical validity, economic value, ethical acceptability, organizational feasibility, and lifecycle governance. - Use AI-HED to preserve the exact intervention identity, evidence cut-off, decision state, scope, safeguards, conditions, and history. - Assess mandatory safeguards independently of comparative scores. - Link monitoring and incidents to explicit transition triggers. - Treat material change and renewal as new decisions requiring proportionate reauthorization. - Preserve authority, continuity, records, and learning through replacement, de-adoption, retirement, and the end of authorization.

## Accountable Adoption Principles

Use the common decision, monitoring, and review record to preserve the final action, conditions, indicators, triggers, review route, and closure requirements.

## Reference Resources

### Key Terms

Table: Key terms for Chapter 12

| Term                  | Working definition and decision relevance                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| Decision architecture | The integrated arrangement connecting evidence, economics, governance, implementation, records, and action.     |
| Algorithmovigilance   | An emerging term for systematic post-deployment evaluation and monitoring of healthcare algorithms.             |
| AI-HED                | The proposed living dossier linking the decision problem, evidence, versions, decisions, and lifecycle actions. |

## Status and Appropriate Use of Selected Instruments

Table: Selected instruments relevant to Chapter 12

| Instrument  | Status or type                        | Appropriate use and limitation                                                                              |
|-------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------|
| AI-HEVG     | Author-developed integrated framework | Structures assessment of problem, data, evidence, economics, ethics, feasibility, and lifecycle governance. |
| AI-HED      | Author-developed decision record      | Maintains a living versioned dossier.                                                                       |
| GATED-AI    | Author-developed process              | Structures decision gates for procurement and approval.                                                     |
| VALIDATE-AI | Author-developed process              | Structures staged implementation.                                                                           |

| Instrument          | Status or type                          | Appropriate use and limitation                                              |
|---------------------|-----------------------------------------|-----------------------------------------------------------------------------|
| SHARE-AI            | Author-developed process                | Structures governed secondary data use.                                     |
| Algorithmovigilance | Emerging scholarly and operational term | Supports post-deployment monitoring; not a standardized regulatory concept. |

## Repository Resource Map

The following resources are indexed to this chapter in the release catalog. Core and worked-case resources form the main route; specialist and computational resources should be selected only when they answer a defined task.

| Resource                                                                                                                                   | Purpose                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-D02 — Lifecycle Decision State, Staged Evidence, and AI-HED Status<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                  | Record the current institutional decision state, exact intervention identity, scope, evidence, conditions, safeguards, triggers, authority, history, and next review.                                      |
| AIHE-D03 — Post-Deployment Monitoring and Algorithmovigilance Cadence<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                    | Define recurring and event-triggered surveillance across technical, workflow, safety, equity, economic, and contractual domains.                                                                           |
| AIHE-D04 — Incident Classification and Reconstruction<br>Core; docx; xlsx; csv; json; md; mmd; svg; png                                    | Reconstruct technical, clinical, workflow, privacy, cybersecurity, equity, communication, and economic incidents consistently.                                                                             |
| AIHE-D05 — Material Change, Reauthorization, Renewal, Replacement, and Retirement Record<br>Core; docx; xlsx; csv; json; md; mmd; svg; png | Assess material change, revalidation, current authorization, renewal, renegotiation, replacement, retirement, continuity, and the end of authorization.                                                    |
| AIHE-CASE12 — Model Update, Incident Response, and Renewal Decision<br>Worked case; md; csv; json; xlsx                                    | Should the organization continue, restrict, roll back, replace, or retire the system after a material update and operational incident?                                                                     |
| AIHE-RG01 — Framework Status and Selection Guide<br>Reference guide; docx; xlsx; csv; json; md                                             | Distinguish legal instruments, standards, guidance, reporting guidelines, appraisal tools, implementation frameworks, and author-developed decision aids; select them according to function and authority. |
| AIHE-AE — AI-HED Completeness and Decision-Quality Review<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png        | Review dossier completeness, confidence, inconsistencies, mandatory safeguards, decision readiness, and remediation.                                                                                       |
| AIHE-E — Applied Case and Decision Review Template<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png               | Apply the book architecture to an institutional case and produce a defensible decision with conditions and lifecycle controls.                                                                             |
| AIHE-EX02 — AI System Inventory and Policy Register<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png              | Maintain a controlled inventory of proposed and deployed AI systems, owners, versions, risk levels, and review dates.                                                                                      |

| Resource                                                                                                                                       | Purpose                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-EX10 — Workflow Change and Training Competency Log<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png              | Document role, workflow, training, competency, downtime, escalation, refresh, and adoption requirements.                                       |
| AIHE-EX11 — Material Change and Renewal Review<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                       | Assess model, data, interface, workflow, vendor, contract, policy, performance, risk, and annual renewal changes.                              |
| AIHE-EX12 — Incident Corrective Action and Reconstruction Log<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png        | Classify and reconstruct technical, clinical, workflow, privacy, cybersecurity, equity, and communication incidents.                           |
| AIHE-EX14 — Final Board Recommendation Pack<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                          | Prepare a decision-ready executive recommendation, options, evidence, economics, residual risk, dissent, conditions, owners, and next review.  |
| AIHE-H — AI Project One-Page Decision Memo<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                           | Summarize the decision problem, comparator, evidence, economics, risks, implementation, conditions, and decision requested.                    |
| AIHE-J — Lifecycle and Future-Proofing Toolkit<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                       | Bundle lifecycle checklists for monitoring, version control, scn planning, policy updates, procurement, and decommissioning.                   |
| AIHE-K — AI-HEVG Assessment Workbook<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                                 | Assess problem relevance, data, validity, economics, ethics, feasibility, lifecycle governance, safeguards, and final decision.                |
| AIHE-P — AI-HED Dossier Template<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                                     | Maintain a versioned evidence, economics, governance, implementation, monitoring, update, and decision record.                                 |
| AIHE-R — VALIDATE-AI Implementation Plan<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                             | Manage implementation as an evidence-producing sequence from problem verification through continuous evaluation.                               |
| AIHE-S — Algorithmovigilance Dashboard, Cadence, and Incident Record<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png | Operate daily technical checks, periodic service and economic review, incident classification, reconstruction, escalation, and annual renewal. |
| AIHE-T — AI De-adoption and Decommissioning Plan<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png                     | Define stopping triggers, fallback, transition, data disposition, communication, contract exit, validation, and post-retirement review.        |
| AIHE-V — Validation Protocol for the Proposed Framework Package<br>Supplemental specialist tool; docx; xlsx; csv; json; md; mmd; svg; png      | Define construct, content, reliability, feasibility, decision, implementation, and prospective validation for proposed frameworks.             |

| Resource                                                                                                                                    | Purpose                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIHE-PY09 — Temporal validation and drift monitoring<br>Python analytics; py; ipynb; xlsx; csv; json; png; md                               | Tracks discrimination, calibration error, score distribution, and feature drift across time periods.                                                                                            |
| AIHE-PY27 — Algorithmovigilance, lifecycle cadence, drift, and incident monitoring<br>Python analytics; py; ipynb; xlsx; csv; json; png; md | Monitors technical, clinical, workflow, equity, economic, and trust indicators against warning and stop thresholds over time.                                                                   |
| AIHE-PY29 — AI-HED Decision-Provenance and Lifecycle-Trigger Audit<br>Python analytics; py; ipynb; csv; json; md                            | Audit intervention identity, evidence versions, current decision state, conditions, overdue review, unresolved triggers, conflicting records, and missing authority.                            |
| AIHE-PY40 — Lifecycle Decision-State and Trigger Review<br>Python analytics; py; ipynb; csv; json; md                                       | Provide an advisory review of the current institutional decision state, conditions, safeguards, transition triggers, authority, and next review without issuing an institutional authorization. |

## Cross-Chapter Resources

The resources below apply across several chapters or support case analysis, methodological application, and publication maintenance. They should be used selectively and kept separate from the core institutional decision record unless they directly inform the decision.

| Resource                                                  | Tier                      | Purpose                                                                                                                    | Formats                   |
|-----------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------|
| AIHE-CASE00 — Common Case-Analysis Method                 | Case-analysis method      | Provide a consistent problem-first, evidence, economic, governance, and lifecycle method for worked and local cases.       | md; docx; xlsx; csv; json |
| AIHE-CASE06B — Financial Recognition and Budget Incidence | Supplementary worked case | Distinguish economic cost, budget incidence, released capacity, cashable savings, revenue, and accounting treatment.       | md; csv; json; xlsx       |
| AIHE-CASE08B — EVSI and Expected Net Benefit of Sampling  | Supplementary worked case | Connect the value of additional evidence with sample size, research cost, implementation delay, and decision consequences. | md; csv; json; xlsx       |
| AIHE-CASE09B — Upstream Model-Provider Change             | Supplementary worked case | Assess continuing disclosure, material change, revalidation, contract response, and reauthorization.                       | md; csv; json; xlsx       |

| Resource                                                | Tier             | Purpose                                                                                                  | Formats                  |
|---------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------|--------------------------|
| AIHE-PY40 — Lifecycle Decision-State and Trigger Review | Python analytics | Provide advisory review of decision state, conditions, safeguards, triggers, authority, and next review. | py; ipynb; csv; json; md |

## Repository Workflow for Institutional Projects

### Suggested Project Folder

```

project-name/
├── 00_source_release/      # Unmodified companion files and citation record
├── 01_decision_specification/ # Problem, comparator, intended use, owners
├── 02_evidence/            # External and local evidence, appraisal, versions
├── 03_analysis/            # Costing, models, notebooks, tests, outputs
├── 04_governance/          # Safeguards, approvals, risk, contracts
├── 05_implementation/      # Training, workflow, pilot, monitoring
├── 06_decisions/           # Signed decisions, conditions, dissent, review dates
└── 07_closure/             # Transition, data closure, decommissioning, lessons

```

### Control Principles

- Keep the published release unchanged and work from a controlled copy.
- Use one accountable document owner and named owners for specialist analyses.
- Link each calculation and statement to a source, date, and version.
- Archive static outputs from interactive notebooks and dashboards when they inform a decision.
- Retain rejected alternatives and dissent so that later review can reconstruct the original choice.
- Use event-triggered review after incidents, material changes, vendor notices, major price changes, new evidence, altered populations, or changed workflows.
- Close a project only after continuity, communication, data retention or deletion, contract closure, asset disposition, and organizational learning are documented.

## Customization and Quality Review

A local adaptation should identify what changed and why. Edits that alter a definition, threshold, formula, score, mandatory safeguard, or decision rule require more scrutiny than edits to layout or local terminology. Before approval, reviewers should confirm that:

1. the decision problem and comparator remain clear;
2. the exact intervention and version are traceable;
3. the data and evidence are current and applicable;
4. assumptions and missing evidence are visible;
5. calculations are reproducible and independently checked in proportion to consequence;
6. subgroup, workload, affordability, continuity, and exit effects are not omitted;
7. mandatory safeguards are assessed independently of any aggregate score;
8. the decision conditions and monitoring triggers are operational; and
9. the completed record is stored, cited, and scheduled for review.

## Versioning and Citation

A recommended citation structure is:

Zouri, M., and Cumpăt, C. M. Artificial Intelligence in Healthcare Management and Economics: Digital Resource Companion. V1.0. 2026. Cite the DOI and repository URL associated with the specific release used.

For a specific resource, add the stable identifier and title:

AIHE-[ID], [Resource title], V1.0, in the Digital Resource Companion, accessed [date], locally adapted by [institution] on [date].

A modified resource should retain the original citation and add a local version statement. Do not imply that a local adaptation has been reviewed or endorsed by the authors unless such review has occurred.

## Glossary

| Term                                 | Definition                                                                                                                                                                                                                                         |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI Health Economics Dossier (AI-HED) | A versioned evidence package linking the decision problem, evidence, economics, governance, implementation, monitoring, material change, and final recommendation                                                                                  |
| AI Health-System Value Chain         | A proposed causal sequence extending from problem and data through model output, decision, action, outcome, resource consequence, and public value                                                                                                 |
| AI-HEVG                              | The proposed Artificial Intelligence Health Economic Value and Governance Framework, integrating problem relevance, data readiness, clinical validity, economic value, ethical acceptability, organizational feasibility, and lifecycle governance |
| algorithmic bias                     | Systematic error or unfairness in artificial intelligence outputs or pathways that disadvantages particular groups, settings, or circumstances                                                                                                     |
| algorithmovigilance                  | Continuing surveillance and governance of the real-world technical, clinical, economic, ethical, equity, workflow, cybersecurity, and trust performance of artificial intelligence-enabled interventions                                           |
| artificial intelligence              | A broad family of computational techniques designed to perform tasks associated with human intelligence, including pattern recognition, prediction, language processing, classification, generation, optimization, and decision support            |
| budget impact analysis               | An assessment of the expected financial consequences of adopting an intervention within a specified budget, population, uptake pattern, and time horizon                                                                                           |

| Term                             | Definition                                                                                                                                                                          |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clinical decision support        | Tools that provide patient-specific information, predictions, alerts, or recommendations to support clinical decisions                                                              |
| corrective and preventive action | A structured process for containing a problem, investigating causes, implementing corrective and preventive measures, and verifying their effectiveness                             |
| cost-effectiveness analysis      | An economic evaluation comparing the incremental costs and outcomes of alternative interventions                                                                                    |
| cost-utility analysis            | A form of cost-effectiveness analysis in which outcomes are commonly expressed as quality-adjusted life years                                                                       |
| data provenance                  | Documented information about the origin, ownership, timing, transformation, version, and lineage of data used in an analysis or artificial intelligence system                      |
| de-adoption                      | A deliberate reduction, restriction, replacement, or cessation of an intervention that no longer provides an acceptable balance of value, evidence, feasibility, and risk           |
| decommissioning                  | The planned technical, contractual, operational, data, and asset activities required to retire a digital or artificial intelligence system safely                                   |
| digital exclusion                | Barriers that prevent people from benefiting from digital health services because of access, literacy, language, disability, income, geography, design, or trust                    |
| digital product passport         | A digital information mechanism used in European sustainable-product policy to provide product information across the value chain, subject to product-specific requirements         |
| GATED-AI                         | A proposed sequential procurement and approval process covering problem, evidence, data, economics, ethics, workflow, contract, monitoring, and exit gates                          |
| generative AI                    | Artificial intelligence systems that generate text, images, code, summaries, recommendations, or other outputs in response to prompts or contextual inputs                          |
| health technology assessment     | A multidisciplinary process that evaluates the clinical, economic, organizational, ethical, and social implications of a health technology to support policy and adoption decisions |

| Term                       | Definition                                                                                                                                                                                                                                     |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| human oversight            | Operational arrangements that enable people to understand, supervise, question, intervene in, override, restrict, or stop an artificial intelligence system                                                                                    |
| interoperability           | The ability of systems and organizations to exchange, interpret, and use information consistently across technical, semantic, organizational, and temporal boundaries                                                                          |
| joint clinical assessment  | A collaborative European Union health technology assessment of relative clinical effectiveness and safety under Regulation (EU) 2021/2282, while national authorities retain responsibility for economic appraisal and reimbursement decisions |
| life-cycle costing         | An approach that estimates costs across acquisition, implementation, use, maintenance, updating, transition, and end-of-life stages within a defined boundary                                                                                  |
| local validation           | Evaluation of a system using data, users, workflows, populations, and operational conditions from the intended local setting before or during controlled deployment                                                                            |
| machine learning           | A subset of artificial intelligence in which a model learns patterns or decision functions from data rather than relying only on explicitly programmed rules                                                                                   |
| material change            | A change in a model, software, data, threshold, intended use, population, workflow, infrastructure, price, vendor arrangement, or governance that may alter performance, risk, value, or obligations                                           |
| material-flow analysis     | A method for tracing the quantities and movement of products, components, or materials into, within, and out of a defined system                                                                                                               |
| model drift                | A decline or change in model performance over time because data, populations, prevalence, workflow, coding, practice, or user behaviour have changed                                                                                           |
| net monetary benefit       | A decision measure that values incremental outcomes at a stated threshold and subtracts incremental cost, facilitating comparison under uncertainty                                                                                            |
| non-compensatory safeguard | A minimum requirement that cannot be offset by favorable scores in other domains, such as patient safety, lawful data use, privacy, cybersecurity, equity, continuity, or accountability                                                       |

| Term                        | Definition                                                                                                                                                                                  |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Public Value AI Scorecard   | A proposed structured summary of clinical, economic, equity, workforce, patient, governance, strategic, and sustainability value                                                            |
| quality-adjusted life year  | A measure combining duration and health-related quality of life, with one unit representing one year in full health                                                                         |
| repairability               | The practical ability to restore function through accessible parts, documentation, diagnostics, skills, service rights, software support, and economically feasible repair                  |
| residual value              | The realizable value remaining in an asset, component, material, licence, or service arrangement at a specified point, net of costs required to recover that value                          |
| return on investment        | A financial measure comparing realized gains or savings with costs; in health systems it should be interpreted alongside clinical, distributional, organizational, and public value         |
| SHARE-AI                    | A proposed process for specifying purpose, harmonizing data, authorizing access, reviewing risks, evaluating value, and auditing implementation of secondary data use                       |
| statistical process control | The use of time-ordered data and control limits to distinguish common-cause variation from signals that warrant investigation or corrective action                                          |
| TCAIO                       | The proposed Total Cost of AI Ownership framework, including acquisition, integration, data, workforce, governance, cybersecurity, monitoring, opportunity, material-change, and exit costs |
| technical debt              | The future cost and risk created when short-term technical choices leave fragile interfaces, undocumented transformations, outdated dependencies, or deferred maintenance                   |
| total cost of ownership     | The full lifecycle cost of acquisition, implementation, integration, operation, monitoring, updating, support, transition, and exit                                                         |
| VALIDATE-AI                 | A proposed implementation pathway comprising Verify, Assess, Local validation, Implement pilot, Determine value, Adapt governance, Transition to scale, and Evaluate continuously           |
| value of information        | An analytic approach estimating the expected benefit of reducing decision uncertainty through additional evidence                                                                           |

## Abbreviations and Acronyms

| Abbreviation  | Meaning                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| ABM           | agent-based model                                                                                                                 |
| AI            | artificial intelligence                                                                                                           |
| AI-HED        | AI Health Economics Dossier                                                                                                       |
| AI-HEVG       | Artificial Intelligence Health Economic Value and Governance Framework                                                            |
| API           | application programming interface                                                                                                 |
| BIA           | budget impact analysis                                                                                                            |
| CAPA          | corrective and preventive action                                                                                                  |
| CEA           | cost-effectiveness analysis                                                                                                       |
| CFIR          | Consolidated Framework for Implementation Research                                                                                |
| CHEERS-AI     | Consolidated Health Economic Evaluation Reporting Standards for interventions using artificial intelligence                       |
| CONSORT-AI    | Consolidated Standards of Reporting Trials–Artificial Intelligence extension                                                      |
| CUA           | cost-utility analysis                                                                                                             |
| DECIDE-AI     | Developmental and Exploratory Clinical Investigations of Decision support systems driven by Artificial Intelligence               |
| DES           | discrete-event simulation                                                                                                         |
| DPP           | digital product passport                                                                                                          |
| EHDS          | European Health Data Space                                                                                                        |
| EHR           | electronic health record                                                                                                          |
| EU            | European Union                                                                                                                    |
| EVPI          | expected value of perfect information                                                                                             |
| EVPPPI        | expected value of partial perfect information                                                                                     |
| FHIR          | Fast Healthcare Interoperability Resources                                                                                        |
| GRADE-CERQual | Grading of Recommendations Assessment, Development and Evaluation–Confidence in the Evidence from Reviews of Qualitative research |

| Abbreviation | Meaning                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------|
| HTA          | health technology assessment                                                                         |
| ICER         | incremental cost-effectiveness ratio                                                                 |
| ISO/IEC      | International Organization for Standardization and International Electrotechnical Commission         |
| JCA          | joint clinical assessment                                                                            |
| LCC          | life-cycle costing                                                                                   |
| MFA          | material-flow analysis                                                                               |
| MLOps        | machine-learning operations                                                                          |
| NASSS        | Nonadoption, Abandonment, Scale-up, Spread, and Sustainability                                       |
| NICE         | National Institute for Health and Care Excellence                                                    |
| NIST         | National Institute of Standards and Technology                                                       |
| NMB          | net monetary benefit                                                                                 |
| OECD         | Organisation for Economic Co-operation and Development                                               |
| OMOP         | Observational Medical Outcomes Partnership common data model                                         |
| PRISMA       | Preferred Reporting Items for Systematic Reviews and Meta-Analyses                                   |
| PROBAST+AI   | Prediction model Risk Of Bias ASsessment Tool plus artificial intelligence                           |
| QALY         | quality-adjusted life year                                                                           |
| ROI          | return on investment                                                                                 |
| SHARE-AI     | Specify, Harmonize, Authorize, Review, Evaluate, and Audit                                           |
| SPC          | statistical process control                                                                          |
| SPIRIT-AI    | Standard Protocol Items: Recommendations for Interventional Trials–Artificial Intelligence extension |
| STARD-AI     | Standards for Reporting Diagnostic Accuracy Studies–Artificial Intelligence extension                |
| TCAIO        | Total Cost of AI Ownership                                                                           |

| Abbreviation | Meaning                                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------|
| TCO          | total cost of ownership                                                                                                              |
| TRIPOD+AI    | Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis plus artificial intelligence         |
| VALIDATE-AI  | Verify, Assess, Local validation, Implement pilot, Determine value, Adapt governance, Transition to scale, and Evaluate continuously |
| WHO          | World Health Organization                                                                                                            |

## Responsible Use

These resources support structured analysis, documentation, education, and institutional deliberation. They do not replace professional judgement or the governance duties of the adopting organization. A completed template does not establish that an intervention is safe, lawful, effective, equitable, affordable, sustainable, or appropriate. The final responsibility is to make evidence, uncertainty, value judgements, ownership, safeguards, and the ability to change course visible enough for an accountable decision.

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